DATA SHEET

TRUEVIEW™ 5725

Multimedia processor for HD PTV and LCD TV

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数据表

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高清电视和液晶电视的多媒体处理器

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INTRODUCTION

The TrueView5725 is a low pin count, low-cost advanced and highly integrated Digital and Analog Video Display Processor providing the key features needed to design HD-READY, Progressive scan or LCD Televisions

TrueView5725 accepts interlaced or progressive video input and graphic input such as NTSC/PAL, 1080i/720p, and SXGA, etc. in both analog and digital channels. It provides a high quality display output. It is capable of advanced de-interlacing, video enhancement, advanced 3D noise reduction, Y/C cross talk suppression and frame rate conversion. It also provides superior video output quality.

It integrates a complete triple 8-bit pipeline ADC with a clock-recovery and sync separation circuit to generate the sampling clock from HSYNC. In applications where the analog outputs are used the 24-bit digital pins can be inputs if the pins are not used for digital output.

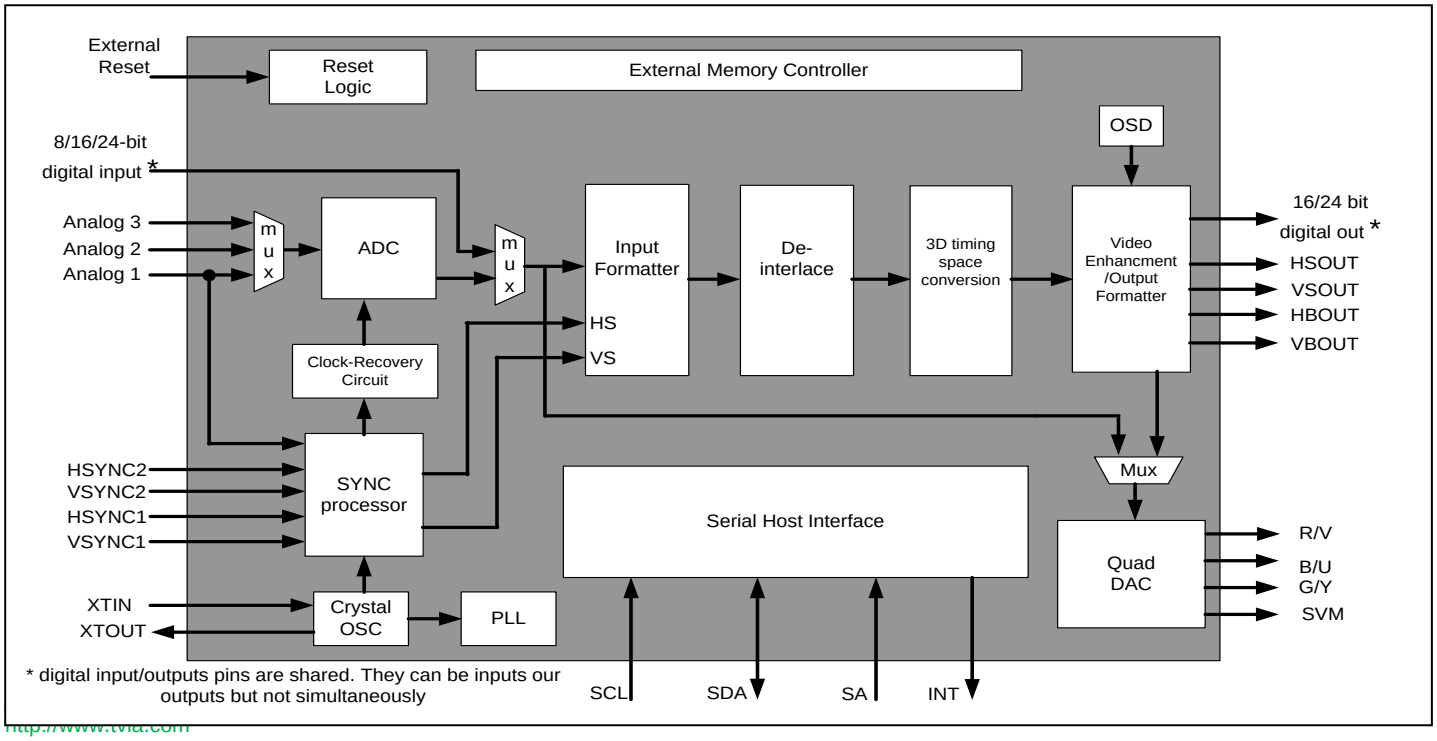
With the high quality video DACs, the video stream is displayed through its de-interlaced RGB/YPbPr outputs.

The I²C host interface enables OEMs to select from many different CPU's in order to meet their system and technical requirements.

OEMs can easily design a very low cost solution, while minimizing their software development time by leveraging on TrueView5725's level of integration and software support.

TrueView5725 provides 3D motion adaptive de-interlacing with diagonal edge detection. It performs high quality line doubling and high-accuracy non-linear motion estimation. It performs pixel based motion detection using two-field buffers. The Video De-interlacing Processor automatically detects noise level and can respond to different noise thresholds. It can also automatically detect input modes such as still images, 3:2 / 2:2 films and provide adaptive processing.

Figure 1: Block Diagram



引言

TrueView5725是一款低引脚数、低成本的先进集成数字模拟视频显示处理器，具备设计高清电视、逐行扫描电视或液晶电视所需的核心功能。

TrueView5725支持隔行扫描或逐行扫描视频输入，以及NTSC/PAL、1080i/720p和 SXGA 等图形输入，支持模拟和数字通道。它能提供高质量的显示输出，具备高级去隔行、视频增强、高级3D降噪、Y/C串扰抑制和帧率转换等功能，同时输出的视频质量也极为出色。

该系统集成了一套完整的三路8位流水线模数转换器(ADC)，并配备时钟恢复与同步分离电路，用于从HSYNC 生成采样时钟。在模拟输出被使用的场景中，若数字引脚未用于数字输出，则可将其作为输入使用。

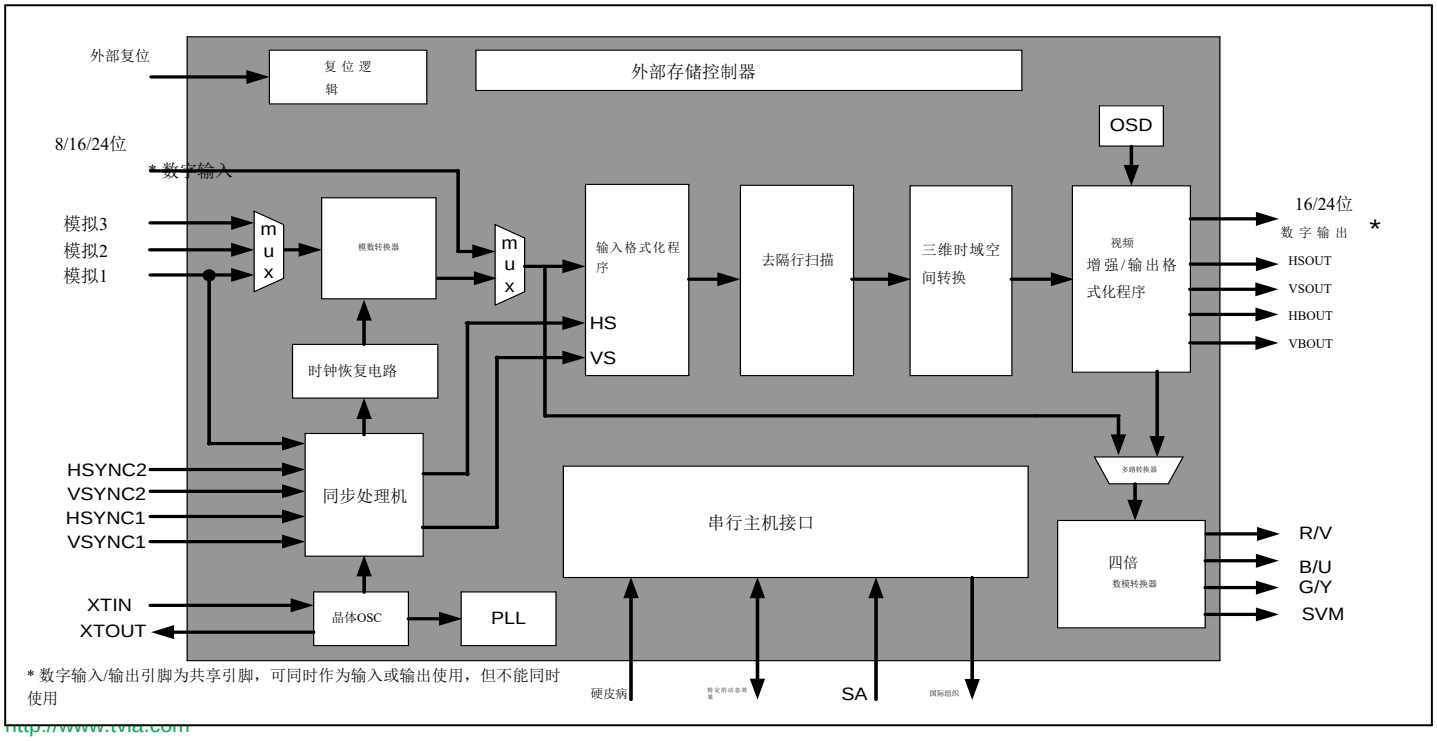
采用高质量视频数模转换器（DAC），视频流通过其去隔行扫描的RGB/YPbPr输出端口呈现。

I²C 主机接口使原始设备制造商能够从多种不同CPU中进行选择，以满足其系统和技术需求。

OEM厂商可借助TrueView5725的集成度与软件支持，轻松设计出低成本解决方案，同时大幅缩短软件开发周期。

TrueView5725采用3D运动自适应去隔行技术，通过斜边检测实现高质量的行倍增和高精度非线性运动估计。该处理器基于像素进行运动检测，采用双场缓冲技术。视频去隔行处理器能自动检测噪声水平，并响应不同噪声阈值。同时，它还能自动识别输入模式（如静态图像、3:2/2:2电影等），并提供相应的自适应处理。

图1：框图



FEATURES

- **Analog Input Formatter**
  - Max three channel analog inputs
  - RGB/YCbCr/YPbPr input
  - Analog input range: 0.5V – 1.0V (p-p)
  - Programmable gain/offset controls
  - DC or AC coupling inputs
  - Internal sync separator to support SOG/SOY
  - SXGA (1280x1024@75Hz) at 135MHz
  - HDTV up to 1080p
  - Macrovision input detection
- **Triple 8-bit ADC**
  - Maximum analog sampling rate up to 162MSPS
- **Clock-recovery Circuit**
  - Programmable phase adjustment cells
  - HSYNC frequency range is from 15KHz to 110KHz
- **Digital Input Formatter**
  - 24bit RGB/YUV input
  - 8/16bit YUV input
  - 8bit 656/601 input
  - NTSC/PAL input
  - 480p, 576p input
  - VGA/SVGA/XVGA input
  - 720p, 1080i, 1080p HD input
  - Support DVI interface
- **Output formatter**
  - 480p, 576p, 720p, 1080i, 1080p
  - Up to SXGA graphic output formats
  - 100/120 interlace double scan
  - 75Hz interlace single scan
  - 50-75Hz scan rate conversion
  - 15~80KHz horizontal frequency
  - 16bit YPbPr digital output with syncs
  - 24bit YPbPr/RGB digital output with syncs
- **De-interlace**
  - HD 1080i support
  - SD NTSC/PAL
  - Direct Edge Correction De-interlace
  - Motion Detection
  - Edge Detection
  - Mode Detection
  - Noise Detection
  - 3:2/2:2 pull-down detection
- **Memory Controller**
  - 2-8Mbyte memory support
  - 16/32bit data access
- **Video enhancement**
  - Black Level Expansion (BLE)
  - White Level Expansion (WLE)
  - Color Transition Improvement (DCTI)
  - Dynamic Range expansion
  - Brightness, Saturation, Contrast, HUE
  - Dynamic 2D Peaking
  - 2D coring
  - 3D noise reduction
  - Scan Velocity Modulation (SVM)
  - 2D Nonlinear scaling
  - Primary Color Enhancement
  - Skin Tone Enhancement
  - Blue Stretch
- **OSD**
  - Simple OSD generator to support component video inputs
- **Host Interface and I/O**
  - Two-wire I²C interface
  - GPIO
- **Quadruple 10bit DAC**
- **Package and Technology**
  - 160-pin QFP 28*28mm
  - 0.18 micron, 3.3V / 1.8V technology

Note: All packages are lead-free and RoHS compliant.

功能

- **模拟输入格式化器**
  - 最多三个模拟输入通道
  - RGB/YCbCr/YPbPr输入
  - 模拟输入范围：0.5V - 1.0V（峰峰值）
  - 可编程增益/偏移控制
  - 直流或交流耦合输入
  - 内部同步分隔符，用于支持SOG/SOY格式
  - SXGA（1280×1024@75Hz）频率135MHz
  - 最高1080p分辨率的高清电视
  - 长焦输入检测

- **三路8位模数转换器**
  - 最高模拟采样率可达162MSPS

- **时钟恢复电路**
  可编程相位调节单元
  - HSYNC 频率范围为15千赫至110千赫

- **数字输入格式化程序**
  - 24位RGB/ YUV 输入
  - 8/16位 YUV 输入
  - 8位656/601输入
  - NTSC/PAL输入
  - 480p、576p输入
  - VGA/SVGA/ XVGA 输入
  - 720p、1080i、1080p高清输入
  - 支持DVI接口

- **输出格式化程序**
  - 480p, 576p, 720p, 1080i, 1080p
  - 支持多达 SXGA 种图形输出格式
  - 100/120隔行双扫描
  - 75Hz隔行扫描
  - 50-75Hz扫描速率转换
  - 15~80千赫水平频率
  - 16位YPbPr数字输出，支持同步信号
  - 24位YPbPr/RGB数字输出，支持同步信号

- **去隔行扫描**
  - 高清1080i支持
    - SD NTSC/PAL
  - 直接边缘校正去隔行
  - 运动检测
  边缘检测
  - 模式检测
  - 噪声检测
  - 3:2/2:2下拉检测

- **存储控制器**
  - 支持2-8MB内存
  - 16/32位数据存取

- **视频增强**
  - 黑电平扩展（BLE）
  - 白电级扩展（WLE）
  - 色彩过渡优化（DCTI）
  动态范围扩展
  - 亮度、饱和度、对比度、HUE
  动态二维峰值
  二维取芯
  - 三维降噪
  - 扫描速度调制（SVM）
  二维非线性缩放
  - 原色增强
  - 皮肤色调提升
  - 蓝色弹力

- **OSD**
  - 简单的OSD生成器，支持组件视频输入

- **主机接口与输入输出**
  双线 I²C 接口
  - GPIO

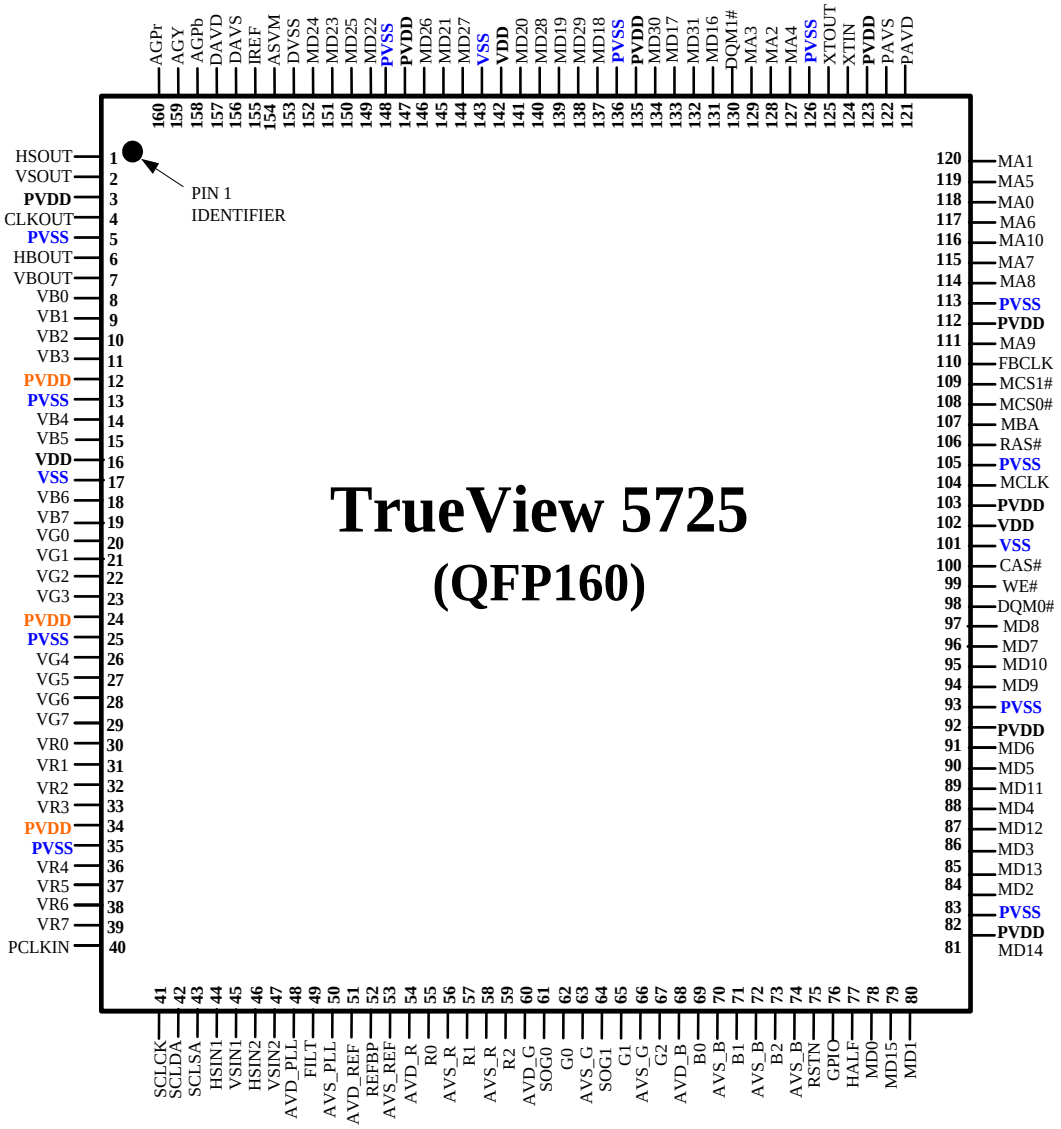
- **四重10位数模转换器**

- **包装与技术**
  - 160-pin QFP 28*28mm
  - 0.18微米，3.3V/1.8V技术

注：所有包装均无铅且符合RoHS标准。

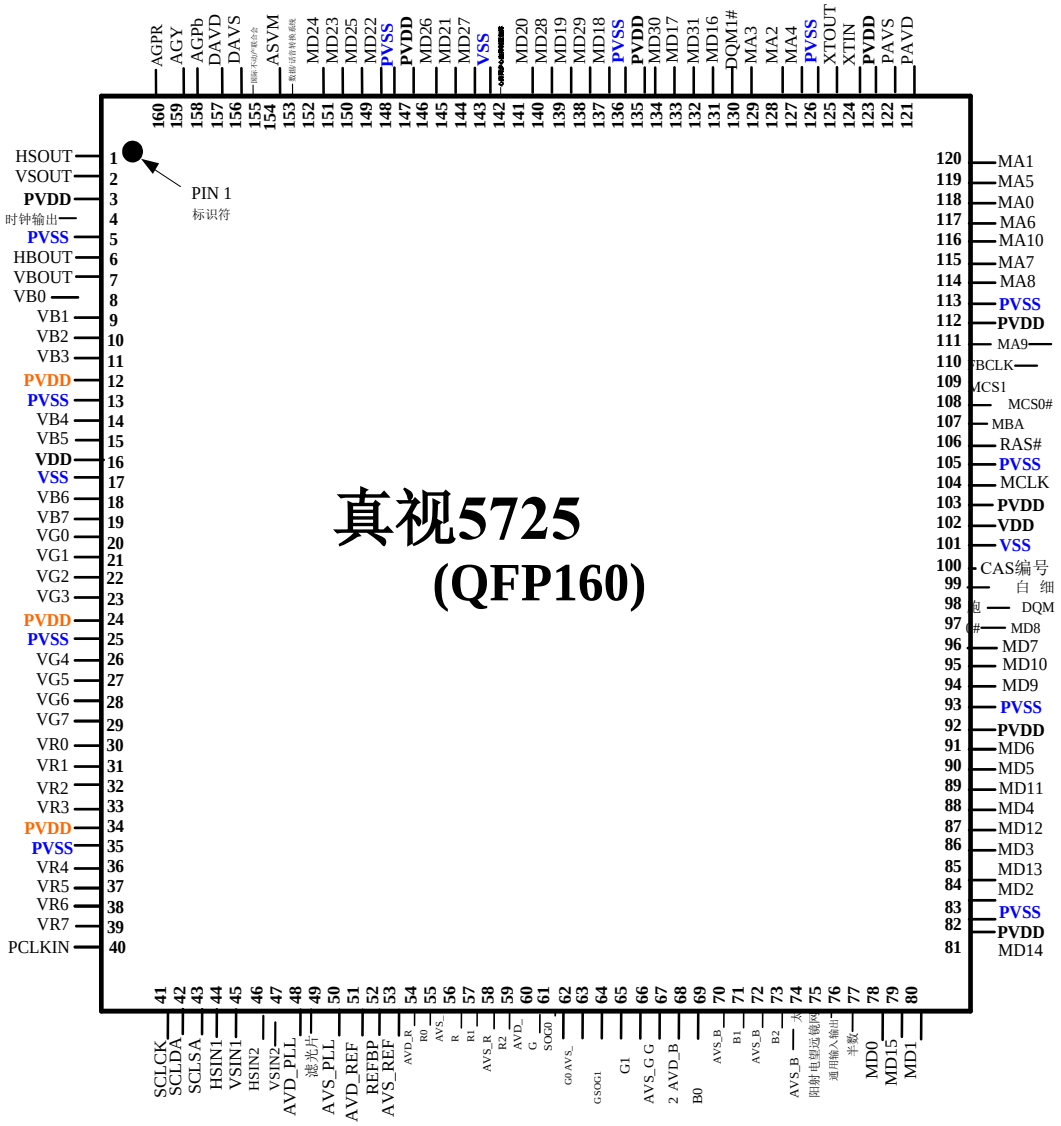
PINOUT DIAGRAM

PIN DIAGRAM FOR QFP160



PINOUT 图

PIN DIAGRAM FOR QFP160



PIN DESCRIPTION

Table 1: Digital Video Input Pins

Pin Name	QFP160 Pin #	Type	Pin Description
PCLKIN	40	IPD	Digital input video clock
HSIN1	44	I	Digital input video H-sync Shared with Analog input video H-sync1
VSIN1	45	I	Digital input video V-sync Shared with Analog input video V-sync1
VG [7:0]	29~26, 23~20	IOPD	Video Green Input data. • G data for 24bit RGB mode. • Y data for 4:4:4 mode • YUV data for 8bit 4:2:2 YUV mode. • Y/G data for 24bit input Shared with digital video output data
VB [7:0]	19~18, 15~14, 11~8	IOPD	Video Blue Input data. • B data for 24bit RGB mode. • U data for 4:4:4 YUV mode. • UV data for 16bit 4:2:2 YUV mode. • U/B data for 24bit input. • Y data for 16bit input. Shared with digital video output data
VR [7:0]	39~36, 33~30	IOPD	Video Red Input data. • R data for 24bit RGB mode. • V data for 4:4:4 mode. • Y data for 16bit 4:2:2 mode. • V/R data for 24bit input • U/V data for 16bit input Shared with digital video output data
CLKOUT	4	IO	Display enable input from DVI/HDMI decoder Shared with video clock output & GPIO bit 7
HSOUT	1	IO	Input H-sync for digital input mode Shared with H-sync output
VSOUT	2	IO	Input V-sync for digital input mode Shared with V-sync output
HALF	77	IOPD	Half tone input Shared with GPIO bit 1

Table 2: Digital Video Output Pins

Pin Name	QFP160 Pin #	Type	Pin Description
CLKOUT	4	IO	Digital video display output clock Shared with display enable input & GPIO bit 7
HSOUT	1	IO	Digital video display output H-sync Shared with H-sync input for digital input mode
VSOUT	2	IO	Digital video display output V-sync Shared with V-sync output for digital input mode

PIN描述

表1：数字视频输入引脚

笔名	QFP160引脚编号	类型	针头描述
PCLKIN	40	插入相位延迟	数字输入视频时钟
HSIN1	44	I	数字输入视频H同步 与模拟输入视频H-sync1共享
VSIN1	45	I	数字输入视频同步 与模拟输入视频V-sync1共享
VG [7:0]	29~26, 23~20	IOPD	视频绿色输入数据。 • 24位RGB模式下的G数据。 • 4:4:4模式的Y数据 • 支持8位4:2:2 YUV 模式的 YUV 数据传输。 • 24位输入的Y/G数据 与数字视频输出数据共享
VB [7:0]	19~18, 15~14, 11~8	IOPD	视频蓝输入数据。 • B数据用于24位RGB模式。 • 支持4:4:4 YUV 模式的视频数据。 • 16位4:2:2 YUV 模式的UV数据。 • 24位输入的U/B数据。 • 16位输入的Y数据。 与数字视频输出数据共享
VR [7:0]	39~36, 33~30	IOPD	视频红色输入数据。 • 24位RGB模式下的R数据。 • 4:4:4模式下的视频数据。 • 16位4:2:2模式下的Y数据。 • 24位输入的V/R数据 • 16位输入的紫外数据 与数字视频输出数据共享
时钟输出	4	IO	显示来自DVI/HDMI解码器的输入信号 与视频时钟输出及 GPIO 位7共享
HSOUT	1	IO	数字输入模式下H同步输入 与H-sync输出共享
VSOUT	2	IO	数字输入模式的V同步输入 与V-sync输出共享
半数	77	IOPD	半音输入 与 GPIO 位1共享

表2：数字视频输出引脚

笔名	QFP160引脚编号	类型	针头描述
时钟输出	4	IO	数字视频显示输出时钟 共享显示启用输入 GPIO 位7
HSOUT	1	IO	数字视频显示输出H同步 与H同步输入共享，支持数字输入模式
VSOUT	2	IO	视频同步输出 与V-sync输出共享以支持数字输入模式

VBOUT	7	IO	Display enable output in LCD application Shared with GPIO bit 6 and V-blank output
VG [7:0]	29~26, 23~20	IOPD	• Y/G data for 24bit digital output. Shared with digital video input data
VB [7:0]	19~18, 15~14, 11~8	IOPD	• UV data for 16bit digital output. • R/V data for 24bit digital output. Shared with digital video input data
VR [7:0]	39~36, 33~30	IOPD	• Y data for 16bit digital output. • B/U data for 24bit digital output. Shared with digital video input data

Table 3: Analog Video input Pins

Pin Name	QFP160 Pin #	Type	Pin Description
HSIN1	44	I	Analog video input H-sync1 Shared with digital video input H-sync
VSIN1	45	I	Analog video input V-sync1 Shared with digital video input V-sync
HSIN2	46	I	Analog video input H-sync2
VSIN2	47	I	Analog video input V-sync2
SOG0	61	AI	Green input 0 for Sync-On-Green sync tip clamping
SOG1	64	AI	Green input 1 for Sync-On-Green sync tip clamping
R0	55	AI	Analog input for Red input channel 0
G0	62	AI	Analog input for Green input channel 0
B0	69	AI	Analog input for Blue input channel 0
R1	57	AI	Analog input for Red input channel 1
G1	65	AI	Analog input for Green input channel 1
B1	71	AI	Analog input for Blue input channel 1
R2	59	AI	Analog input for Red input channel 2
G2	67	AI	Analog input for Green input channel 2
B2	73	AI	Analog input for Blue input channel 2
REFBP	52	AI	Internal reference bypass
AVD_R	54	AP	Analog power (3.3V)
AVS_R	56, 58	AG	Analog ground
AVD_G	60	AP	Analog power (3.3V)
AVS_G	63, 66	AG	Analog ground
AVD_B	68	AP	Analog power (3.3V)
AVS_B	70, 72, 74	AG	Analog ground
AVD_REF	51	AP	Analog power (3.3V)
AVS_REF	53	AG	Analog ground
AVD_PLL	48	AP	Analog power (3.3V) for PLLAD
FILT	49	AI	Connection for External Filters components for PLLAD
AVS_PLL	50	AG	Analog ground for PLLAD

VBOUT	7	IO	LCD应用中显示输出端口 与 GPIO 位6及V-空白输出共享
VG [7:0]	29~26, 23~20	IOPD	• 24位数字输出的Y/G数据。 与数字视频输入数据共享
VB [7:0]	19~18, 15~14, 11~8	IOPD	• 16位数字输出的紫外线数据。 • 用于24位数字输出的R/V数据。 与数字视频输入数据共享
VR [7:0]	39~36, 33~30	IOPD	• 16位数字输出的Y数据。 • 24位数字输出的B/U数据。 与数字视频输入数据共享

表3：模拟视频输入引脚

笔名	QFP160引脚编号	类型	针头描述
HSIN1	44	I	模拟视频输入H同步1 与数字视频输入H同步共享
VSIN1	45	I	模拟视频输入V同步1 与数字视频输入V同步共享
HSIN2	46	I	模拟视频输入H同步2
VSIN2	47	I	模拟视频输入V同步2
SOG0	61	人工智能	绿色输入0用于同步-绿色同步提示夹持
SOG1	64	人工智能	绿色输入1用于同步-绿色同步尖端夹持
R0	55	人工智能	红色输入通道0的模拟输入
G0	62	人工智能	绿色输入通道0的模拟输入
B0	69	人工智能	蓝色输入通道0的模拟输入
R1	57	人工智能	红色输入通道1的模拟输入
G1	65	人工智能	绿色输入通道1的模拟输入
B1	71	人工智能	蓝色输入通道1的模拟输入
R2	59	人工智能	红色输入通道2的模拟输入
G2	67	人工智能	绿色输入通道2的模拟输入
B2	73	人工智能	蓝色输入通道2的模拟输入
REFBP	52	人工智能	内部参考绕过
AVD_R	54	AP	模拟电源（3.3V）
AVS_R	56, 58	总检察长	模拟地
AVD_G	60	AP	模拟电源（3.3V）
AVS_G	63, 66	总检察长	模拟地
AVD_B	68	AP	模拟电源（3.3V）
AVS_B	70, 72, 74	总检察长	模拟地
AVD_REF	51	AP	模拟电源（3.3V）
AVS_REF	53	总检察长	模拟地
AVD_PLL	48	AP	PLLAD 模拟电源（3.3V）
滤光片	49	人工智能	PLLAD 外部过滤器组件连接
AVS_PLL	50	总检察长	PLLAD 模拟地

Table 4: Analog Video Output Interface Pins

Pin Name	QFP160 Pin #	Type	Pin Description
HSOUT	1	IO	Analog video display output H-sync Shared with H-sync input for digital input mode
VSOUT	2	IO	Analog video display output V-sync Shared with V-sync output for digital input mode
HBOUT	6	IO	Analog video display output H-Blank Shared with GPIO bit 5
VBOUT	7	IO	Analog video display output V-Blank Shared with GPIO bit 6 and DE output for digital video output
AGPb	158	AO	Analog Blue/Pb output
AGY	159	AO	Analog Green/Y output
AGPr	160	AO	Analog Red/Pr output
ASVM	154	AO	Analog SVM output
IREF	155	AI	Full-scale adjust resistor
DVSS	153	AG	Big current GND
DAVS	156	AG	Analog ground for DAC
DAVD	157	AP	Analog power for DAC

Table 5: Clock Generation Pins

Pin Name	QFP160 Pin #	Type	Pin Description
XTOUT	125	O	External crystal output.
XTIN	124	I	External crystal input.
PAVD	121	AP	Analog power (3.3V) for PLL648
PAVS	122	AG	Analog ground for PLL648

Table 6: System Interface Pins

Pin Name	QFP160 Pin #	Type	Pin Description
SCLSA	43	IOPU	Serial bus slave address selection, Shared with GPIO bit 2
SCLCK	41	I	Serial bus clock
SCLDA	42	IO	Serial bus data
RSTN	75	I	External asynchronous reset, low active
GPIO	76	IOPD	GPIO bit 0 Shared with Interrupt Output, low active

表4：模拟视频输出接口引脚

笔名	QFP160引脚编号	类型	针头描述
HSOUT	1	IO	模拟视频显示输出H同步 与H同步输入共享，支持数字输入模式
VSOUT	2	IO	模拟视频显示输出V同步 与V-sync输出共享以支持数字输入模式
HBOUT	6	IO	模拟视频显示输出H-空白 与 GPIO 位5共享
VBOUT	7	IO	模拟视频显示输出V-Blank 与 GPIO 位6及DE输出共享，用于数字视频输出
AGPb	158	安哥拉	模拟蓝/铅输出
AGY	159	安哥拉	模拟绿色/黄色输出
AGPR	160	安哥拉	模拟红/Pr输出
ASVM	154	安哥拉	模拟 SVM 输出
国际不动产联合会	155	人工智能	全量程调节电阻器
数据/话音转换系统	153	总检察长	大电流 GND
DAVS	156	总检察长	数模转换器模拟地
DAVD	157	AP	DAC模拟电源

表5：时钟生成引脚

笔名	QFP160引脚编号	类型	针头描述
XTOUT	125	O	外晶体输出
XTIN	124	I	外部晶体输入
PAVD	121	AP	PLL648模拟电源（3.3V）
PAVS	122	总检察长	PLL648模拟地

表6：系统接口引脚

笔名	QFP160引脚编号	类型	针头描述
SCLSA	43	输入输出处理装置	串行总线从站地址选择 与 GPIO 位2共享
SCLCK	41	I	串行总线时钟
SCLDA	42	IO	串行总线数据
太阳射电望远镜网	75	I	外部异步复位，低电平激活
通用输入输出	76	IOPD	GPIO 位0 与中断输出共享，低电平有效

Table 7: SDRAM Interface Pins

Pin Name	QFP160 Pin #	Type	Pin Description
MD31	132	IOPD	Memory Data Bus [31:0]
MD30	134	IOPD	
MD29	138	IOPD	
MD28	140	IOPD	
MD27	144	IOPD	
MD26	146	IOPD	
MD25	150	IOPD	
MD24	152	IOPD	
MD23	151	IOPD	
MD22	149	IOPD	
MD21	145	IOPD	
MD20	141	IOPD	
MD19	139	IOPD	
MD18	137	IOPD	
MD17	133	IOPD	
MD16	131	IOPD	
MD15	79	IOPD	
MD14	81	IOPD	
MD13	85	IOPD	
MD12	87	IOPD	
MD11	89	IOPD	
MD10	95	IOPD	
MD9	94	IOPD	
MD8	97	IOPD	
MD7	96	IOPD	
MD6	91	IOPD	
MD5	90	IOPD	
MD4	88	IOPD	
MD3	86	IOPD	
MD2	84	IOPD	
MD1	80	IOPD	
MD0	78	IOPD	
DQM0#	98	O	Memory data qualify signal 0
DQM1#	130	O	Memory data qualify signal 1
MCLK	104	O	SDRAM clock
WE#	99	O	Write enable control for SDRAM
RAS#	106	O	Row address strobe
CAS#	100	O	Column address strobe
FBCLK	110	IOPD	Feed back clock for SDRAM Chip Selection 2 for 6MByte external memory
MBA	107	IOPD	SDRAM bank select Shared with GPIO bit 3

表7：SDRAM接口引脚

笔名	QFP160引脚编号	类型	针头描述
MD31	132	IOPD	存储器数据总线[31:0]
MD30	134	IOPD	
MD29	138	IOPD	
MD28	140	IOPD	
MD27	144	IOPD	
MD26	146	IOPD	
MD25	150	IOPD	
MD24	152	IOPD	
MD23	151	IOPD	
MD22	149	IOPD	
MD21	145	IOPD	
MD20	141	IOPD	
MD19	139	IOPD	
MD18	137	IOPD	
MD17	133	IOPD	
MD16	131	IOPD	
MD15	79	IOPD	
MD14	81	IOPD	
MD13	85	IOPD	
MD12	87	IOPD	
MD11	89	IOPD	
MD10	95	IOPD	
MD9	94	IOPD	
MD8	97	IOPD	
MD7	96	IOPD	
MD6	91	IOPD	
MD5	90	IOPD	
MD4	88	IOPD	
MD3	86	IOPD	
MD2	84	IOPD	
MD1	80	IOPD	
MD0	78	IOPD	
DQM0#	98	O	存储器数据合格信号0
DQM1#	130	O	存储器数据合格信号1
MCLK	104	O	存储器数据随机存取时钟
白细胞	99	O	写入SDRAM启用控制
RAS#	106	O	行地址选通脉冲
CAS编号	100	O	列地址选通
FBCLK	110	IOPD	SDRAM反馈时钟 6M字节外存芯片选择（二）
MBA	107	IOPD	sdram bank select 与 GPIO 位3共享

Pin Name	QFP160 Pin #	Type	Pin Description
MCS1#	109	IOPD	Memory chip Selection 1, And shared with GPIO bit 4.
MCS0#	108	IOPD	Memory chip Selection 0
MA10	116	IOPD	**Memory address bus [10:0]**
MA9	111	IOPD	
MA8	114	IOPD	
MA7	115	IOPD	
MA6	117	IOPD	
MA5	119	IOPD	
MA4	127	IOPD	
MA3	129	IOPD	
MA2	128	IOPD	
MA1	120	IOPD	
MA0	118	IOPD	

Table 8: Digital Power and Ground Pins

Pin Name	QFP160 Pin #	Type	Pin Description
VSS	17,101,143	DG	Core Power GND
VDD	16,102,142	DP	1.8V Core Power
PVDD	3, 12, 24, 34, 82, 92, 103, 112, 123, 135, 147,	DP	**3.3V I/O Power**
PVSS	5, 13, 25, 35, 83, 93, 105, 113, 126, 136, 148	DG	I/O GND

Pin types include the following:

- I Digital Input
- IPD Digital Input with pull-down
- O Digital Output
- OPD Digital Output with pull-down
- IO Digital Bi-directional (input/output)
- IOPU Digital Bi-directional (input/output) with pull-up
- IOPD Digital Bi-directional (input/output) with pull-down
- AI Analog Input
- AO Analog Output
- DP Digital Power
- DG Digital Ground
- AP Analog Power
- AG Analog Ground

笔名	QFP160引脚编号	类型	针头描述
MCS1	109	IOPD	存储器芯片选择1， 并与 GPIO 第4位共享。
MCS0#	108	IOPD	存储器芯片选择0
MA10	116	IOPD	存储器地址总线[10:0]
MA9	111	IOPD	
MA8	114	IOPD	
MA7	115	IOPD	
MA6	117	IOPD	
MA5	119	IOPD	
MA4	127	IOPD	
MA3	129	IOPD	
MA2	128	IOPD	
MA1	120	IOPD	
MA0	118	IOPD	

表8：数字电源与接地引脚

笔名	QFP160引脚编号	类型	针头描述
VSS	17,101,143	DG	核心电源 GND
心房同步心室抑制型起搏	16,102,142	DP	1.8V核心电源
PVDD	3, 12, 24, 34, 82, 92, 103, 112, 123, 135, 147,	DP	**3.3V I/O电源**
PVSS	5, 13, 25, 35, 83, 93, 105, 113, 126, 136, 148	DG	I/O GND

针头类型包括以下几种：

- I　数字输入
- 插入相位延迟　下拉式数字输入
- O　数字输出
- OPD　下拉式数字输出
- IO　数字双向（输入/输出）
- 输入输出处理装置　数字双向（输入/输出）上拉
- IOPD　数字双向（输入/输出）下拉式
- 人工智能　模拟输入
- 安哥拉　模拟输出
- DP　数字电源
- DG　数字地
- AP　模拟电源
- 总检察长　模拟地

Table 9: Numerical Pin List

Pin Name	QFP160 Pin #	Pin Name	QFP160 Pin #	Pin Name	QFP160 Pin #	Pin Name	QFP160 Pin #
HSOUT	1	SCLCK	41	MD14	81	PAVD	121
VSOUT	2	SCLDA	42	PVDD	82	PAVS	122
PVDD	3	SCLSA	43			PVDD	123
PVDD		HSIN1	44	PVSS	83		
CLKOUT	4	VSIN1	45	MD2	84	XTIN	124
PVSS	5	HSIN2	46	MD13	85	XTOUT	125
PVSS		VSIN2	47	MD3	86	PVSS	126
HBOUT	6	AVD_PLL	48	MD12	87	PVSS	
VBOUT	7	AVD_PLL		MD4	88	MA4	127
VB0	8	FILT	49	MD11	89	MA2	128
VB1	9	AVS_PLL	50	MD5	90	MA3	129
VB2	10	AVD_REF	51	MD6	91	DQM1	130
VB3	11	REFBP	52	PVDD	92	MD16	131
PVDD	12	AVS_REF	53	PVSS	93	MD31	132
PVSS	13	AVD_R	54	MD9	94	MD17	133
VB4	14	AVD_R		MD10	95	MD30	134
VB5	15	R0	55	MD7	96	PVDD	135
VDD	16	AVS_R	56	MD8	97	PVSS	136
VDD		R1	57	DQM0	98	MD18	137
VSS	17	AVS_R	58	WE#	99	MD29	138
VB6	18	R2	59	CAS#	100	MD19	139
VB7	19	AVD_G	60	VSS	101	MD28	140
VG0	20	AVD_G		VDD	102	MD20	141
VG1	21	SOG0	61	VDD		VDD	142
VG2	22	G0	62	PVDD	103	VDD	
VG3	23	AVS_G	63	PVDD		VSS	143
PVDD	24	SOG1	64	MCLK	104	VSS	
PVSS	25	G1	65	PVSS	105	MD27	144
VG4	26	AVS_G	66	PVSS		MD21	145
VG5	27	G2	67	RAS#	106	MD26	146
VG6	28	AVD_B	68	MBA	107	PVDD	147
VG7	29	AVD_B		MCS0#	108	PVSS	148
VR0	30	B0	69	MCS1#	109	MD22	149
VR1	31	AVS_B	70	FBCLK	110	MD25	150
VR2	32	B1	71	MA9	111	MD23	151
VR3	33	AVS_B	72	PVDD	112	MD24	152
PVDD	34	B2	73	PVSS	113	DVSS	153
PVDD		AVS_B	74	MA8	114	ASVM	154
PVSS	35	RSTN	75	MA7	115	IREF	155
VR4	36	GPIO	76	MA10	116	DAVS	156
VR5	37	HALF	77	MA6	117	DAVD	157
VR6	38	MD0	78	MA0	118	AGPb	158
VR7	39	MD15	79	MA5	119	AGY	159
PCLKIN	40	MD1	80	MA1	120	AGPr	160

表9：数字引脚列表

笔名	QFP160引脚编号	笔名	QFP160引脚编号	笔名	QFP160引脚编号	笔名	QFP160引脚编号
HSOUT	1	SCLCK	41	MD14	81	PAVD	121
VSOUT	2	SCLDA	42	PVDD	82	PAVS	122
PVDD	3	SCLSA	43			PVDD	123
PVDD		HSIN1	44	PVSS	83		
时钟输出	4	VSIN1	45	MD2	84	XTIN	124
PVSS	5	HSIN2	46	MD13	85	XTOUT	125
PVSS		VSIN2	47	MD3	86	PVSS	126
HBOUT	6	AVD_PLL	48	MD12	87	PVSS	
VBOUT	7	AVD_PLL		MD4	88	MA4	127
VB0	8	滤光片	49	MD11	89	MA2	128
VB1	9	AVS_PLL	50	MD5	90	MA3	129
VB2	10	AVD_REF	51	MD6	91	DQM1	130
VB3	11	REFBP	52	PVDD	92	MD16	131
PVDD	12	AVS_REF	53	PVSS	93	MD31	132
PVSS	13	AVD_R	54	MD9	94	MD17	133
VB4	14	AVD_R		MD10	95	MD30	134
VB5	15	R0	55	MD7	96	PVDD	135
心房同步心室抑制型起搏	16	AVS_R	56	MD8	97	PVSS	136
心房同步心室抑制型起搏		R1	57	DQM0	98	MD18	137
VSS	17	AVS_R	58	白细胞	99	MD29	138
VB6	18	R2	59	CAS编号	100	MD19	139
VB7	19	AVD_G	60	VSS	101	MD28	140
VG0	20	AVD_G		心房同步心室抑制型起搏	102	MD20	141
VG1	21	SOG0	61	心房同步心室抑制型起搏		心房同步心室抑制型起搏	142
VG2	22	G0	62	PVDD	103	心房同步心室抑制型起搏	
VG3	23	AVS_G	63	PVDD		VSS	143
PVDD	24	SOG1	64	MCLK	104	VSS	
PVSS	25	G1	65	PVSS	105	MD27	144
VG4	26	AVS_G	66	PVSS		MD21	145
VG5	27	G2	67	RAS#	106	MD26	146
VG6	28	AVD_B	68	MBA	107	PVDD	147
VG7	29	AVD_B		MCS0#	108	PVSS	148
VR0	30	B0	69	MCS1	109	MD22	149
VR1	31	AVS_B	70	FBCLK	110	MD25	150
VR2	32	B1	71	MA9	111	MD23	151
VR3	33	AVS_B	72	PVDD	112	MD24	152
PVDD	34	B2	73	PVSS	113	数据/话音转换系统	153
PVDD		AVS_B	74	MA8	114	ASVM	154
PVSS	35	太阳射电望远镜网	75	MA7	115	国际不动产联合会	155
VR4	36	通用输入输出	76	MA10	116	DAVS	156
VR5	37	半数	77	MA6	117	DAVD	157
VR6	38	MD0	78	MA0	118	AGPb	158
VR7	39	MD15	79	MA5	119	AGY	159
PCLKIN	40	MD1	80	MA1	120	AGPR	160

FUNCTIONAL DESCRIPTION

The TrueView 5725 TV Display Processor consists of a number of functional blocks which are described below.

ADC

The ADC can accommodate input signals with inputs ranging from 0.5V to 1.0V full scale. The full-scale range is set in three 8-bit registers (Red Gain, Green Gain, and Blue Gain).
The ADC offset control shifts the input source a DC level, there are three 7-bit registers (Red Offset, Green Offset, Blue Offset) available to independently control each channel, The offset controls provide a ± 63 LSB adjustment range. If the input range is 1.0V, the adjustable range is ±0.25V. Every step is 4mV (0.25 / 63 = 0.004V). If the input range is 0.5V, the adjustable range is ±0.125V. Every step is 2mV (0.125 / 63 = 0.002V).
The ADC has three input data channels, two SOG input channels and two separate H/V-sync input channels. It also generates an over-sampling clock from H-sync with a frequency range from lower than 10MHz up to 162Mhz.

INPUT FORMATTER

The Input Formatter accepts video data in several formats.

- 24-bit YCbCr or RGB
- 4:4:4 or 4:2:2
- 8 bit ITU-R BT.656
- 8 or 16 or 24 bit ITU-R BT.601
- NTSC or PAL
- 24 bit VGA (800*525@60Hz), SVGA, XGA and SXGA
- 1080i, 720p, 1080p, HD

The True View 5725 can automatically detect all supported monitor and TV modes including NTSC, PAL, VGA, SVGA, XGA, SXGA and HD modes and their supported refresh rate.

DE-INTERLACER

The TrueView5725 Motion Adaptive De-interlacer automatically determines the type of incoming video content such as film, static interlaced or moving interlaced video. Different algorithms are applied for each content type. The de-interlacer produces a progressive scan video output. Noise which otherwise would be seen as motion can be removed during motion processing. It also uses an edge detection algorithm that enables the smoothing of jagged edges that could occur in the de-interlacing process. Several different detection angles can be programmed.

FILM MODE

The De-interlace Processor can detect video that has originated from film and detects 3:2 or 2:2 pull down. If film originated video is detected the two fields from the original frame are combined into one progressive frame.

Thirty frames per second interlaced video can be output as 60 or 72Hz progressive frames.

Twenty-five frames per second interlaced video can be output as 50 or 75Hz progressive frames, or 100Hz-interlaced fields.

The de-interlacer continuously monitors the incoming video to detect any changes to the pull-down sequence. If a change is detected the deinterlacer quickly adapts its de-interlacing algorithm to compensate.

3D TIME – SPACE CONVERTER

The 3D Time – Space converter is used to alter the size of the picture depending on the output resolution. It contains an Up-Down scaler and Rate converter.

VIDEO PROCESSING

The Video Enhancer is a high-quality programmable processor that brings out details and color in the video.

Output Formatter

The Output formatter formats the output to give the required levels for RGB or YPbPr. It contains a 2X interpolator to increase sharpness that results from a smaller aperture.

功能描述

TrueView 5725电视显示处理器由多个功能模块组成，具体如下所述。

模数转换器

该模数转换器(ADC)可处理输入信号范围为0.5V至1.0V满量程的输入信号。满量程范围通过三个8位寄存器（红色增益、绿色增益和蓝色增益）进行设置。
ADC偏移控制可将输入源偏移一个直流电平，系统提供三个7位寄存器（红偏移、绿偏移、蓝偏移）用于独立控制各通道。偏移控制提供±63 LSB 的调节范围。当输入范围为1.0V时，可调范围为±0.25V，每步4mV（0.25/63=0.004V）；当输入范围为0.5V时，可调范围为±0.125V，每步2mV（0.125/63=0.002V）。
该模数转换器(ADC)具有三个输入数据通道，两个同步输出通道(SOG)和两个独立的行/垂直同步通道(H/V-sync)。它还能从行同步信号中生成过采样时钟，频率范围从低于10MHz到162MHz。

我 NPUT F 或物质

输入格式化器可处理多种格式的视频数据。

- 24位YCbCr或RGB
- 4:4:4 或 4:2:2
- 8位ITU-R BT.656
- 8位、16位或24位ITU-R BT.601
- NTSC或PAL
- 24位VGA (800*525@60Hz) 、SVGA、 XGA 和 SXGA
- 1080i、720p、1080p、高清

TrueView5725可自动识别所有支持的显示器和电视模式，包括NTSC、PAL、VGA、SVGA、 XGA 、 SXGA和HD模式及其对应支持的刷新率。

D E-交织器

TrueView5725运动自适应去隔行扫描技术可自动识别图像类型输入的视频内容包括电影、静态隔行扫描或动态隔行扫描视频。针对不同内容类型采用不同的算法处理。去隔行扫描器可生成逐行扫描视频输出。在运动处理过程中，原本可能被误判为运动的噪声可被消除。该设备还采用边缘检测算法，可有效平滑去隔行扫描过程中可能出现的锯齿状边缘。系统支持多种检测角度的编程设置。

FILM MODE

该去隔行处理器能够检测源自胶片的视频，并识别3:2或2:2的下拉模式。若检测到胶片源视频，系统会将原始帧的两个场合并为一个逐行扫描帧。

每秒三十帧的隔行扫描视频可输出为60或72Hz的逐行扫描帧。

每秒25帧的隔行扫描视频可输出为50或75Hz的逐行扫描帧，或100Hz的隔行扫描场。

去隔行扫描器持续监控输入视频，以检测下拉序列的任何变化。若检测到变化，该去隔行扫描器将迅速调整其去隔行扫描算法以进行补偿。

3D T IME - S PACE C 转换器

3D时空转换器用于根据输出分辨率改变图像尺寸，其包含上下缩放器和速率转换器。

V 想象 P 处理

视频增强器是一款高品质可编程处理器，能显著提升视频细节与色彩表现。

输出格式化程序

输出格式化器可将输出格式化为RGB或YPbPr所需级别。其内置2倍插值器，可增强因光圈缩小而产生的锐度。

Transient Improvement

Digital Chroma Transient improvement (DCTI) improves video by replacing slower edges of the video with edges that have steeper rise and fall times. DCTI turns sloped or sinusoidal waveforms into rectangular or square waveforms with the same duty cycles and peak-to-peak amplitude. It improves the color transitions of vertical objects and reduces color smearing introduced by a video decoder.

Black & white Level Expansion

The black and white-level expander enhances the contrast of the picture. The luminance signal is modified with an adjustable, non-linear function. Dark areas of the picture are made blacker, while bright areas of the picture are made whiter.

2D Peaking

Especially if the input signals are decoded composite signals and a notch filter has been used for luminance/color separation, it is necessary to improve the luminance/color frequency characteristics. The TrueView5725 2D peaking process uses edge enhancement to improve the Y and C sharpness in the vertical and horizontal directions. For Y, there is 2D peaking and for UV, there is only vertical peaking.

Skin tone correction

Skin tone correction is used to alter skin color to make it look more natural.

Non-linear saturation

Non-linear saturation is used to enhance the saturation of some colors without affecting other colors. Both the Y and UV parameters of the range of colors to be enhanced are programmable.

Brightness and Contrast

Brightness and contrast can be programmed by adjusting the offset and gain of the video signal. Contrast is adjusted by multiplying the luminance by a constant. Brightness is adjusted by adding or subtracting a constant, from the luminance value.

Saturation

Saturation can be changed by changing the UV gain of the color signal

Hue

The Hue of the color can be changed by rotating the UV vectors in either direction.

Background Noise Reduction

Background noise reduction is a process that uses a digital filtering algorithm on the digital image data to reduce the amount of random noise such as RF noise or comb filter artifacts in composite video sources etc. It is 3D motion adaptive and pixel based

Scan Velocity Modulation

The Scan Velocity Modulation (SVM) output is used to modulate the horizontal CRT scan timing. It has the effect of sharpening the transitions from dark to light and light to dark which results in a clearer sharper picture.

ANALOG AND DIGITAL OUTPUT

The Analog Display Port generates the analog RGB or YPbPr with triple Digital-to-Analog Converters (DACs). The DAC's have 10-bit quality. The analog RGB or YPbPr output is generated in synchronization with H and V timing signals. The 16-bit YCbCr digital output is ITU-R BT.601 compatible with a 1 X clock. The output comes from the 3D time-space converter block and is in 4:2:2 format.

MEMORY CONTROLLER

The internal Memory Controller supports the addressing and control of up to 8MB of external SDRAM. The external SDRAM memory buffer is used to store video fields and motion data. The Memory Controller also supports frame rate up-conversion with different input and output refresh rates.

PLL AND OSCILLATOR

The TrueView 5725 integrates a PLL to generate the MCLK to the Memory interface and the VCLK to the display. Only an external crystal is required to be connected between the XTIN and XTOUT pins. On power-up, the PLL is initialized to provide a 108 MHz MCLK and a 27 MHz VCLK when a 27MHz reference is used. Alternatively, the MCLK and VCLK can be driven directly by external clocks.

HOST BUS INTERFACE

Access to the TrueView 5725 registers is provided by an I²C 2-wire serial bus interface. The Interface

一过性改善

数字色度瞬态改善（DCTI）技术通过用上升和下降时间更陡峭的边缘替代视频中较慢的边缘来优化视频质量。该技术将斜坡或正弦波形转换为矩形或方波形，保持相同的占空比和峰峰值幅度。它能改善垂直物体的色彩过渡，并减少视频解码器导致的色彩模糊。

黑白级扩展

该黑白级扩展器可增强图像对比度。亮度信号通过可调非线性函数进行修正，使图像暗部区域更趋黑色，亮部区域更趋白色。

二维峰值

当输入信号为解码后的复合信号且已采用陷波滤波器进行亮度/色度分离时，需优化其亮度/色度频率特性。TrueView5725 2D峰值处理技术通过边缘增强技术，可提升Y和C通道在垂直与水平方向的锐度。其中Y通道采用二维峰值处理，而UV通道仅进行垂直方向峰值处理。

肤色矫正

肤色校正用于改变肤色以使其看起来更自然。

非线性饱和

非线性饱和度用于增强某些颜色的饱和度而不影响其他颜色。待增强颜色范围的Y参数和UV参数均可编程设置。

亮度与对比度

亮度和对比度可通过调节视频信号的偏移量和增益进行编程。对比度通过将亮度值乘以一个常数来调整，而亮度则通过从亮度值中加减一个常数来调节。

饱和

可通过调节彩色信号的紫外增益来改变饱和度

色彩

通过旋转可调节颜色的色调

双向紫外线载体

背景噪声降低

背景降噪是一种通过对数字图像数据应用数字滤波算法，以减少随机噪声（如射频噪声或复合视频源中的梳状滤波伪影）的处理过程。该技术具有三维运动自适应性和基于像素的特性。

扫描速度调制

扫描速度调制（SVM）输出用于调节CRT水平扫描时序，可使明暗过渡更加锐利，从而获得更清晰锐利的图像。

A NALOG 和 D IGITAL O UTPUT

模拟显示端口通过三个数字-模拟转换器(DAC)生成RGB或YPbPr模拟信号，转换器具有10位精度，模拟信号与H、V同步，16位YCbCr数字输出符合ITU-R BT.601标准，输出时钟为1X，输出信号来自3D时-空转换器，输出格式为4:2:2。

M 内存 C 控制器

内部存储控制器支持对高达8MB的外部SDRAM进行寻址和控制。外部SDRAM存储缓冲区用于存储视频场和运动数据。该存储控制器还支持不同输入和输出刷新率的帧率上变频。

P 柱和 O 振荡器

TrueView5725集成有生成 MCLK 至存储器接口和 VCLK 至显示器的 PLL 。仅需在 XTIN 和 XTOUT 引脚之间连接一个外部晶体。上电时， PLL 初始化为提供108MHz MCLK ，当使用27MHz参考时则提供27MHz VCLK 。此外， MCLK 和 VCLK 也可由外部时钟直接驱动。

H OST B US I nterface

TrueView 5725寄存器的访问通过 I²C 双线串行总线接口实现。该接口

supports standard and fast modes, up to 400kbits/S. Only slave mode is supported in the TrueView 5725 TV Display Processor.

The slave address is hardware selectable.

Table 10: I²C Slave Addresses

TrueView 5725 I²C slave addresses	
Read	Write
2F	2E
AF	AE

OSD

The TrueView 5725 OSD engine is a simple eight icon, hardware graphic engine. It consists of eight 13x13 hard-wired OSD ROM locations containing the most common TV OSD symbols. Brightness, contrast, hue, sound, up/down adjustment, left/right adjustment, horizontal size adjustment, and vertical size adjustment.

Figure 2: OSD Icons

One of eight foreground and background colors can be selected and they can be zoomed in the horizontal or vertical direction up to eight times horizontally and four times vertically.

CLOCK RECOVERY CIRCUITS

The clock recovery circuit generates the ADC sampling clock and its over-sampling clock up to 162Mhz.The hsync signal can also be recovered from the PLL even when the input clock reference is a composite sync signal. This is achieved by sending a COAST high signal to the PLL to disable the PLL from tracking the input reference and to maintain the PLL output clock frequency if the input reference is not a regular period hsync signal. FS, KS and ICP registers are used to optimize the PLL performance for each specific video format. The CKOS register is used to control the over-sampling ratio. The following table gives the possible over-sampling cases. The over sampling ratio is determined by the relationship of the KS and CKOS register values.

Table 11: Over sampling ratio

KS	CKOS	OSR
00	00	1
01	00	2
	01	1
10	00	4
	01	2
	10	1

支持标准模式和高速模式，最高传输速率可达400千比特/秒。TrueView 5725电视显示处理器仅支持从属模式。

从设备地址为硬件可选。

表10： I²C 从属地址

TrueView 5725 I²C 从属地址	
阅读	写入
2F	2E
心房颤动	AE

OSD

TrueView 5725 OSD引擎是一款简易的八图标硬件图形引擎，由八个13x13的硬连线OSD ROM位组成，包含电视OSD最常用的符号，包括亮度、对比度、色调、声音、上下调节、左右调节、水平尺寸调节和垂直尺寸调节。

图2：OSD图标

可从八种前景与背景颜色中选择一种，并支持水平方向最多八倍、垂直方向最多四倍的缩放。

C 锁定 R 恢复 C 电路

时钟恢复电路可生成ADC采样时钟及其最高达162MHz的过采样时钟。即使输入时钟参考为复合同步信号，仍能从 PLL 中恢复出hsync信号。通过向 PLL 发送高电平信号来实现这一功能，该信号可使 PLL 禁用跟踪输入参考信号，并在输入参考信号不是标准周期的水平同步信号时保持 PLL 输出时钟频率。

FS、KS和ICP寄存器用于优化针对每种特定视频格式的 PLL 性能。 CKOS 寄存器用于控制过采样率。下表列出了可能的过采样情况。过采样率由以下因素决定。

KS与 CKOS 寄存器值之间的关系

表11：过采样比率

KS	CKOS	OSR
00	00	1
01	00	2
	01	1
10	00	4
	01	2
	10	1

APPLICATION NOTES

ADC PIN CONNECTION

The analog RGB signals are connected to the TrueView5725 as shown below:

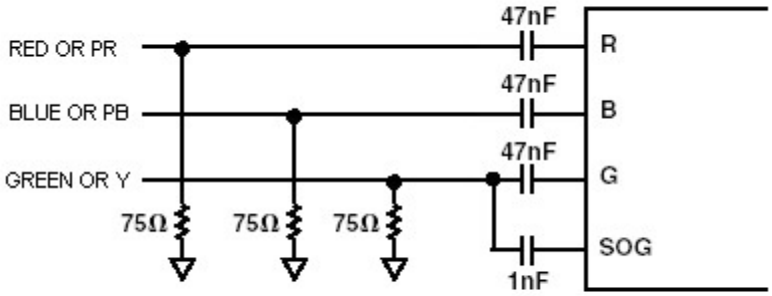


Figure 3: Example ADC Signal Connection

应用说明

ADC P IN C onnection

模拟RGB信号连接至TrueView5725，如下图所示：

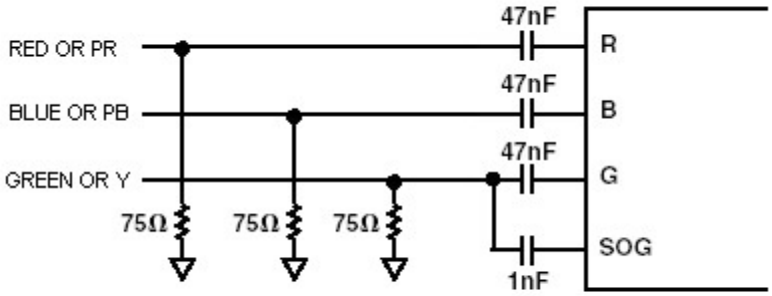


图3：ADC信号连接示例

CRYSTAL PIN

1. EXTERNAL

XTIN and XTOUT are the input and output, respectively, of an inverting amplifier, which can be configured for use as an on-chip oscillator, as shown below. A 27 MHz crystal is preferred. In addition to an external crystal, the oscillator requires two external capacitors and an external feedback resistor.

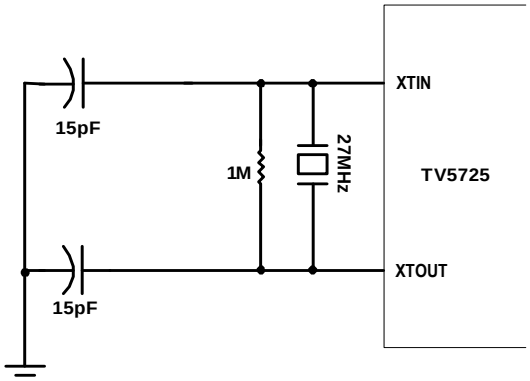


Figure 4: Crystal Connection Diagram

To drive the device from an external clock source, XTIN should be driven, while XTOUT floats, as shown below.

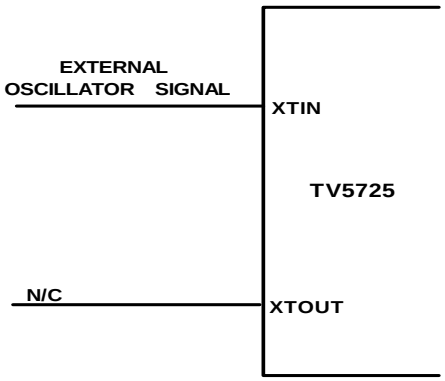


Figure 5: External Clock Drive Configuration

CRYSTAL PIN

1. 外源性的

XTIN 和 XTOUT 分别是反相放大器的输入和输出，该放大器可配置为片上振荡器，如下所示。优选使用27MHz晶体。除外部晶体外，该振荡器还需两个外部电容器和一个外部反馈电阻。

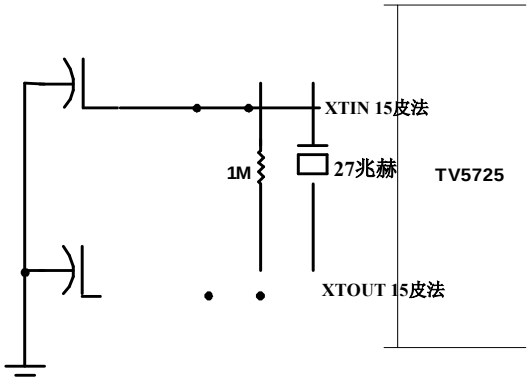


图4：晶体连接示意图

为使设备从外部时钟源驱动，应驱动 XTIN ，同时 XTOUT 保持浮动状态，如下图所示。

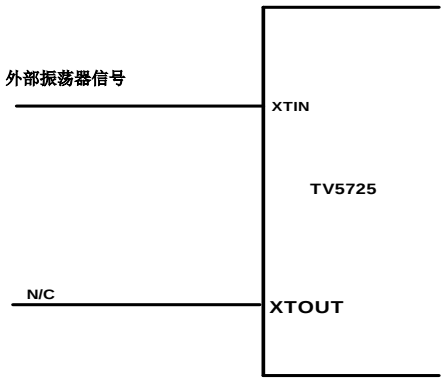


图5：外部时钟驱动配置

2. INTERNAL

C[2:0] are register bits, for changing the clock driver drive capability. The default value is 0 as shown below. SMT_EN_O is the register bit to select between a TTL or Schmitt trigger input buffer.

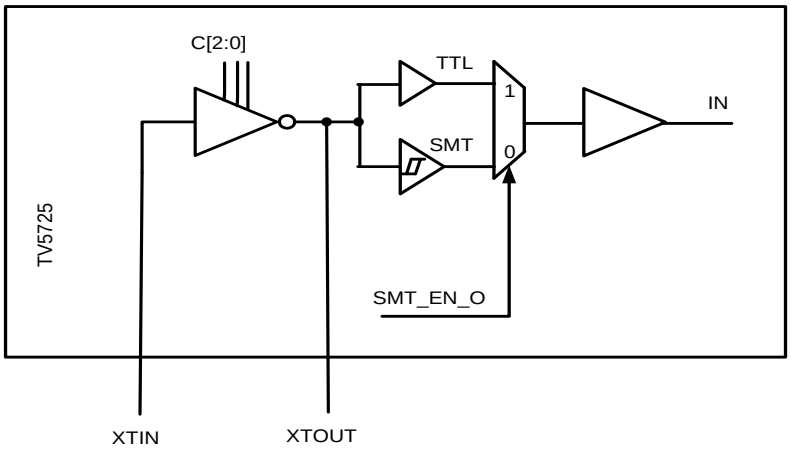


Figure 6: Internal Driver Configuration

2. 内部

C[2:0]是寄存器位，用于调整时钟驱动器的驱动能力。默认值为0，如下所示。SMT_EN_O是用于选择 TTL 或Schmitt触发输入缓冲器的寄存器位。

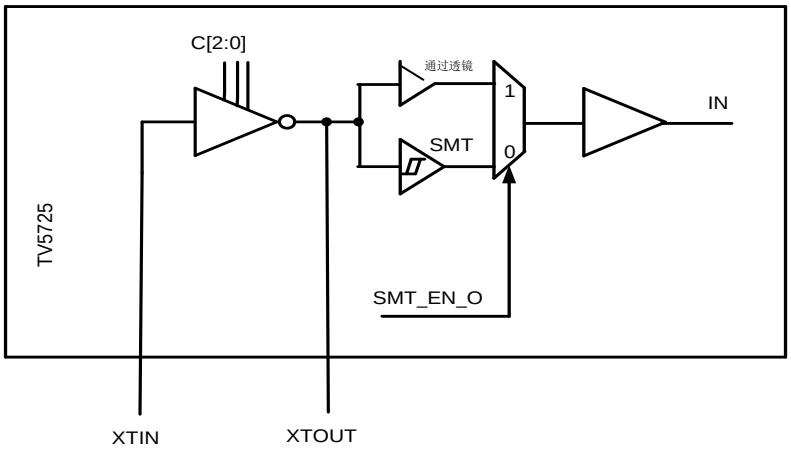


图6：内部驱动器配置

VIDEO PORT USAGE

The TrueView 5725 has both analog and digital video input ports.

For analog input port:

- Three ADC input channels 0, 1, 2
- Two separate sync h/v or composite sync for ADC channels 0, 1, 2
- Two SOG inputs for ADC input channels 0, 1

For analog output port:

- One set of RGBS, H/V-sync, H/V-blank outputs
- It can output sync-on-Y YPbPr or sync-on-green RGB signal

For digital input port:

- One digital 24-bit input port
- 8-bit 601/656 YUV, 16-bit YUV, 24-bit YUV/RGB input
- De-in (display enable input) signal

For digital output port:

- Shared with 24-bit digital input port
- 16-bit YUV, 24-bit YUV/RGB.
- De-out (display enable output) signal
- H/V-sync and pixel clock
- H/V Blank <----------Note: H/V blank could be used for letter-box mode and pillar-box mode

The following pages show the port usage and corresponding register programming:

视频端口使用情况

TrueView 5725配备模拟和数字视频输入接口。

对于模拟输入端口：

- 三个ADC输入通道0、1、2
- ADC通道0、1、2的两个独立同步高/低电平或复合同步信号
- ADC输入通道0、1的两个SOG输入

对于模拟输出端口：

- 一组 RGBS ，H/V同步，H/V空白输出
- 可输出同步-Y YPbPr或同步-绿色RGB信号

对于数字输入端口：

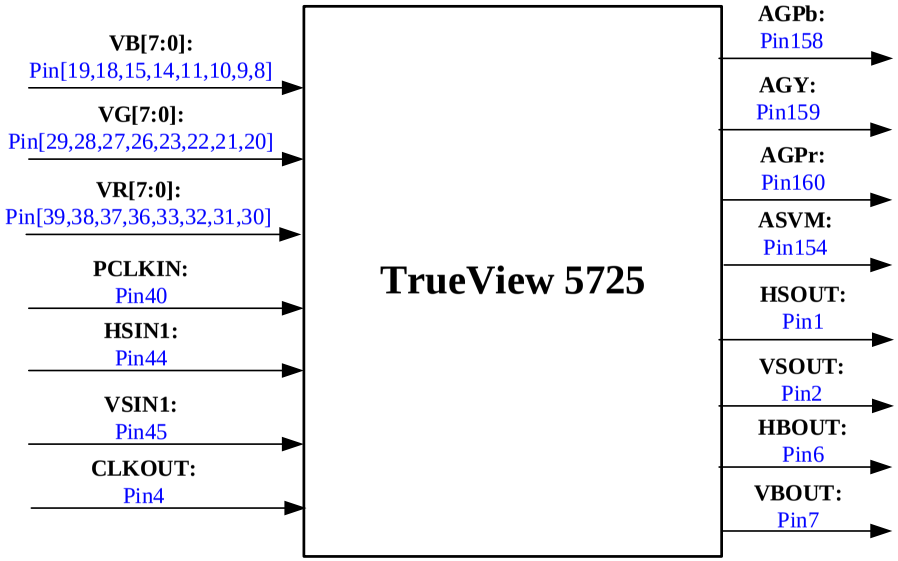
- 一个数字24位输入端口
- 8位601/656型 YUV ，16位 YUV ，24位 YUV /RGB输入
- 显示使能输入信号

对于数字输出端口：

- 支持24位数字输入端口
- 16位 YUV ，24位 YUV /RGB。
- 显示使能输出信号
- H/V同步与像素时钟
- H/V空白 <----------注：H/V空白可用于信封模式和柱状模式

以下页面展示了端口使用情况及对应的寄存器编程：

System Solution 1: Digital 24-bit YUV/RGB Input with Analog Output Mode



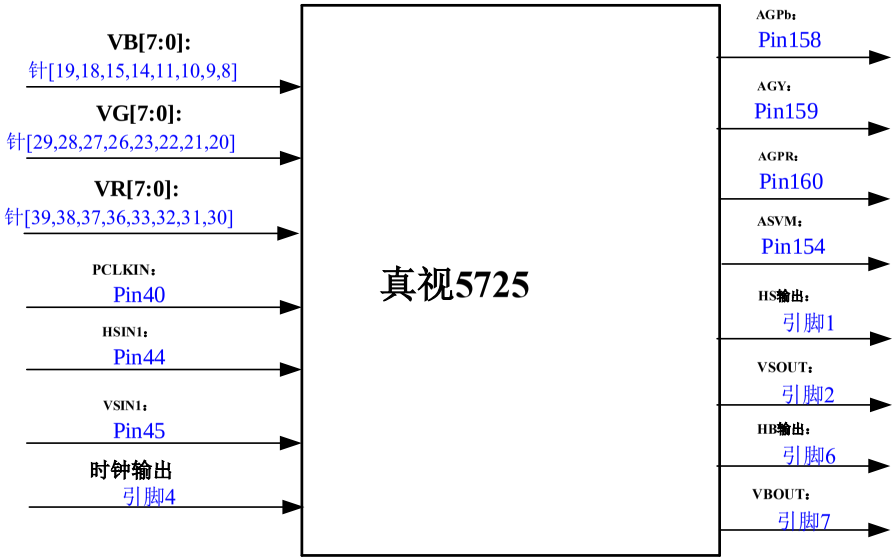
Input Pin	Description	Output Pin	Description
VB[7:0]	Digital Blue/U data input	AGPb	Analog Blue/Pb output
VG[7:0]	Digital Green/Y data input	AGY	Analog Green/Y output
VR[7:0]	Digital Red/V data input	AGPr	Analog Red/Pr output
PCLKIN	Pixel clock input	ASVM	Analog SVM output
HSIN1	Video H-sync input	HSOUT	Video H-sync output
VSIN1	Video V-sync input	VSOUT	Video V-sync output
CLKOUT	DE input from DVI/HDMI	HBOUT	Video H-blank output
		VBOUT	Video V-blank output

Figure 7: Digital 24-bit YUV/RGB Input with Analog Output Mode

5725 Key Register Setting		
Register Name	**Address**	**Value**
pad_bout_en	Reg_S0_48[0]	1'b0
pad_bin_enz	Reg_S0_48[1]	1'b0
pad_rout_en	Reg_S0_48[2]	1'b0
pad_rin_enz	Reg_S0_48[3]	1'b0
pad_gout_en	Reg_S0_48[4]	1'b0
pad_gin_enz	Reg_S0_48[5]	1'b0
pad_ckout_enz	Reg_S0_49[1]	1'b1
pad_sync_out_enz	Reg_S0_49[2]	1'b0
pad_blk_out_enz	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
if_sel_adc_sync	Reg_S1_28[2]	1'b0
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	X
if_sel24bit	Reg_S1_01[7]	1'b1

Note: "X" means either "0" or "1" is OK.

系统解决方案1：数字24位 YUV /RGB输入，支持模拟输出模式



输入引脚	描述	输出引脚	描述
VB[7:0]	数字蓝/蓝数据输入	AGPb	模拟蓝/铅输出
VG[7:0]	数字绿色/Y数据输入	AGY	模拟绿色/黄色输出
VR[7:0]	数字红/绿数据输入	AGPR	模拟红/Pr输出
PCLKIN	时钟输入	ASVM	模拟 SVM 输出
HSIN1	视频H同步输入	HSOUT	视频H同步输出
VSIN1	视频同步输入	VSOUT	视频同步输出
时钟输出	DVI/HDMI输入的DE信号	HBOUT	视频H空白输出
		VBOUT	视频空白输出

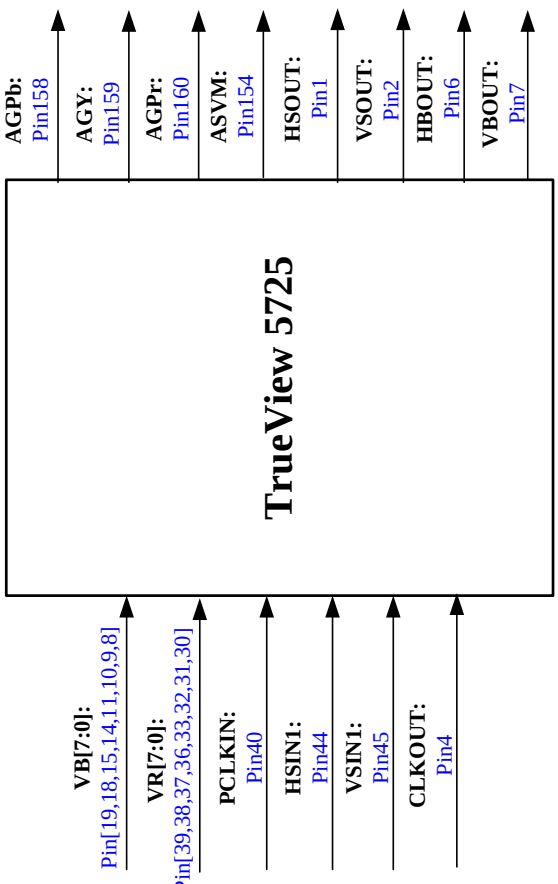
图7：24位 YUV /RGB输入模式（模拟输出）

5725 密钥寄存器设置		
注册名称	地址	值
垫脚	Reg_S0_48[0]	1'b0
酶的截断	Reg_S0_48[1]	1'b0
并行输出通道	Reg_S0_48[2]	1'b0
垫片	Reg_S0_48[3]	1'b0
垫式关节炎	Reg_S0_48[4]	1'b0
垫片	Reg_S0_48[5]	1'b0
包装出货	Reg_S0_49[1]	1'b1
同步酶的粘附	Reg_S0_49[2]	1'b0
包块输出结束信号	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
选择ADC同步	Reg_S1_28[2]	1'b0
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	X
if_sel24bit	Reg_S1_01[7]	1'b1

注：“X”表示“0”或“1”均可。

Figure 8: Digital 16-bit YUV 4:2:2 Input with Analog Output Mode

System Solution 2: Digital 16-bit YUV 4:2:2 Input with Analog Output Mode

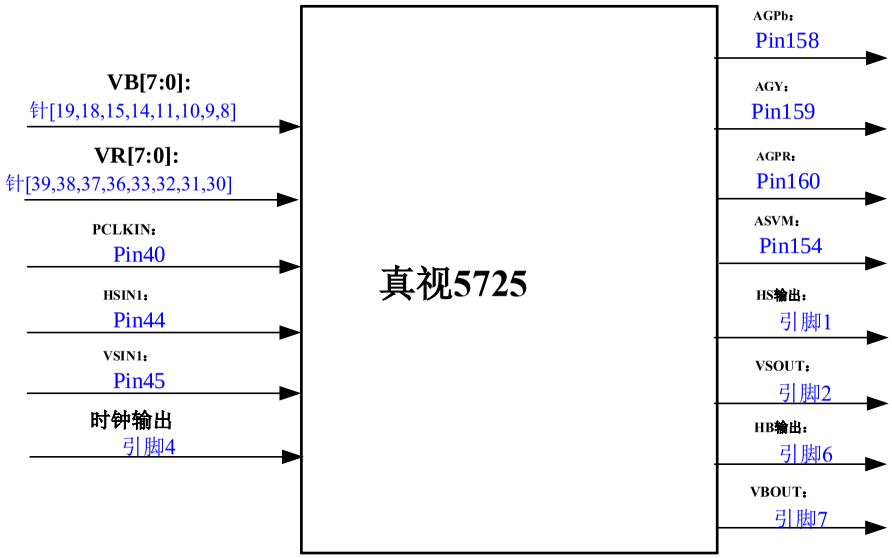


Input Pin	Description	Output Pin	Description
VB[7:0]	Digital UV data input	AGPb	Analog Blue/Pb output
VR[7:0]	Digital Y data input	AGY	Analog Green/Y output
PCLKIN	Pixel clock input	AGPr	Analog Red/Pr output
HSIN1	Video H-sync input	ASVM	Analog SVM output
VSIN1	Video V-sync input	HSOUT	Video H-sync output
CLKOUT	DE input from DVI/HDMI	VSOUT	Video V-sync output
		HBOUT	Video H-blank output
		VBOUT	Video V-blank output

5725 Key Register Setting		
Register Name	**Address**	**Value**
pad_bout_en	Reg_S0_48[0]	1'b0
pad_bin_enz	Reg_S0_48[1]	1'b0
pad_rout_en	Reg_S0_48[2]	1'b0
pad_rin_enz	Reg_S0_48[3]	1'b0
pad_gout_en	Reg_S0_48[4]	X
pad_gin_enz	Reg_S0_48[5]	X
pad_ckout_enz	Reg_S0_49[1]	1'b1
pad_sync_out_enz	Reg_S0_49[2]	1'b1
pad_blk_out_enz	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
if_sel_adc_sync	Reg_S1_28[2]	1'b0
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	1'b1
if_sel24bit	Reg_S1_01[7]	1'b0

Note: "X" means either "0" or "1" is OK.

系统解决方案2：数字16位 YUV 4:2:2输入，支持模拟输出模式



输入引脚	描述	输出引脚	描述
VB[7:0]	数字紫外数据输入	AGPb	模拟蓝/铅输出
VR[7:0]	数字Y数据输入	AGY	模拟绿色/黄色输出
PCLKIN	时钟输入	AGPR	模拟红/Pr输出
HSIN1	视频H同步输入	ASVM	模拟 SVM 输出
VSIN1	视频同步输入	HSOUT	视频H同步输出
时钟输出	DVI/HDMI输入的DE信号	VSOUT	视频同步输出
		HBOUT	视频H空白输出
		VBOUT	视频空白输出

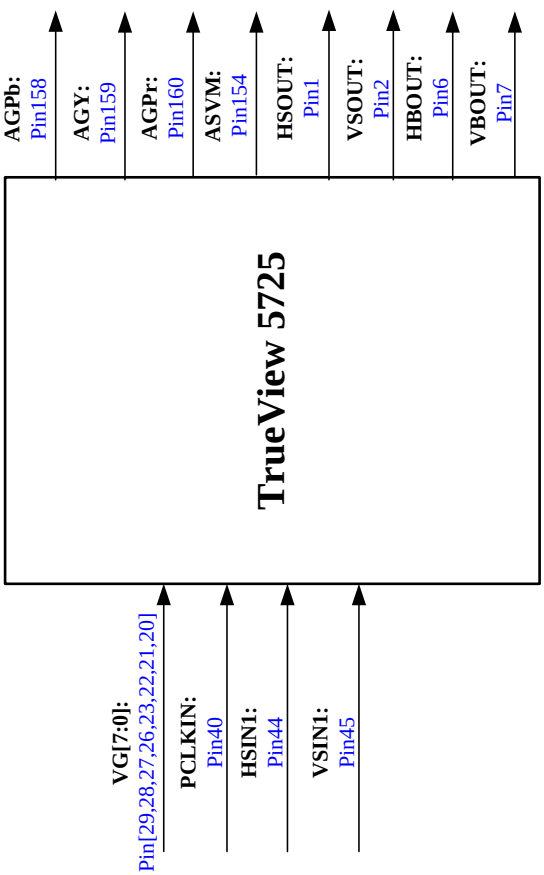
图8：16位数字输入 YUV 4:2:2模拟输出模式

5725 密钥寄存器设置		
注册名称	地址	值
垫脚	Reg_S0_48[0]	1'b0
酶的截断	Reg_S0_48[1]	1'b0
并行输出通道	Reg_S0_48[2]	1'b0
垫片	Reg_S0_48[3]	1'b0
垫式关节炎	Reg_S0_48[4]	X
垫片	Reg_S0_48[5]	X
包装出货	Reg_S0_49[1]	1'b1
同步酶的粘附	Reg_S0_49[2]	1'b0
包块输出结束信号	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
选择ADC同步	Reg_S1_28[2]	1'b0
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	1'b1
if_sel24bit	Reg_S1_01[7]	1'b0

注：“X”表示“0”或“1”均可。

Figure 9: Digital 8-bit 601/656 4:2:2 YUV Input with Analog Output Mode

System Solution 3: Digital 8-bit 601/656 4:2:2 YUV Input with Analog Output Mode



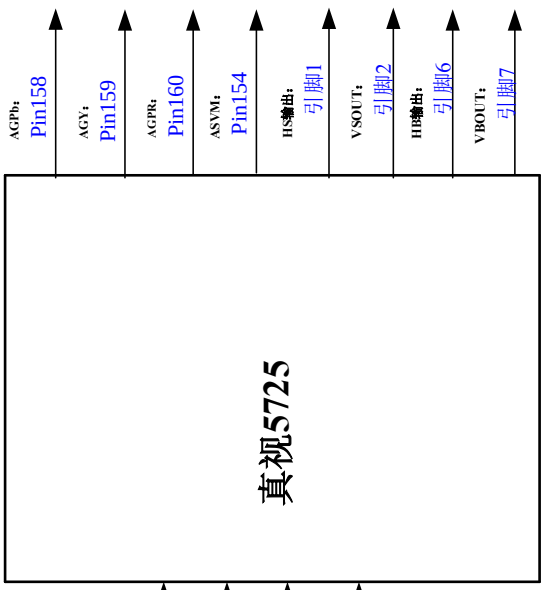
5725 Key Register Setting		
Register Name	**Address**	**Value**
pad_bout_en	Reg_S0_48[0]	X
pad_bin_enz	Reg_S0_48[1]	X
pad_rout_en	Reg_S0_48[2]	X
pad_rin_enz	Reg_S0_48[3]	X
pad_gout_en	Reg_S0_48[4]	1'b0
pad_gin_enz	Reg_S0_48[5]	1'b0
pad_ckout_enz	Reg_S0_49[1]	X
pad_sync_out_enz	Reg_S0_49[2]	1'b0
pad_blk_out_enz	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
if_sel_adc_sync	Reg_S1_28[2]	1'b0
if_sel_656	Reg_S1_00[3]	Note*
if_sel16bit	Reg_S1_00[4]	1'b0
if_sel24bit	Reg_S1_01[7]	1'b0

Note: "X" means either "0" or "1" is OK.
Note* : if 656 input , set it to 1;
if 601 input , set it to 0.

Input Pin	Description	Output Pin	Description
VG[7:0]	Digital YUV data input	AGPb	Analog Blue/Pb output
PCLKIN	Pixel clock input	AGY	Analog Green/Y output
HSIN1	Video H-sync input	AGPr	Analog Red/Pr output
VSIN1	Video V-sync input	ASVM	Analog SVM output
		HSOUT	Video H-sync output
		VSOUT	Video V-sync output
		HBOUT	Video H-blank output
		VBOUT	Video V-blank output

图9：数字输入8位601/656 4:2:2 YUV 模拟输出模式

系统解决方案3：数字8位601/656 4:2:2 YUV 输入，支持模拟输出模式

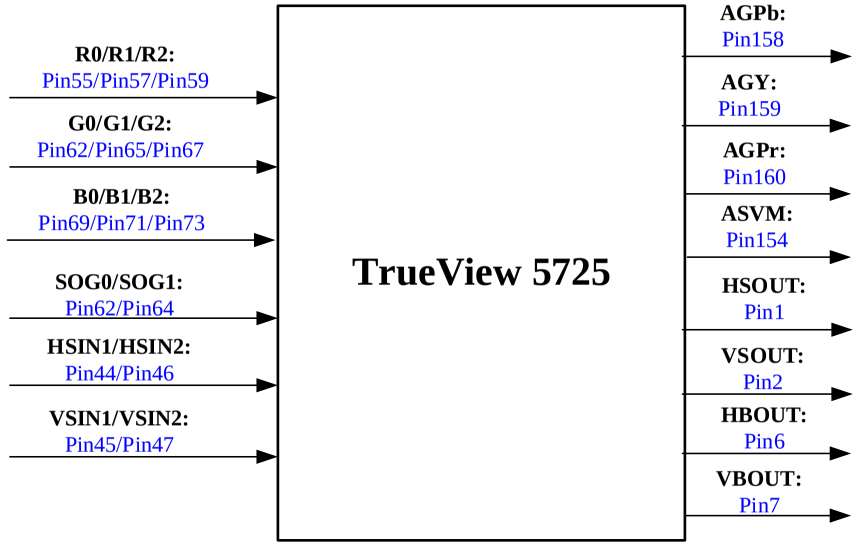


5725 密钥寄存器设置		
注册名称	地址	值
垫脚	Reg_S0_48[0]	X
酶的截断	Reg_S0_48[1]	X
并行输出通道	Reg_S0_48[2]	X
垫片	Reg_S0_48[3]	X
垫式关节炎	Reg_S0_48[4]	1'b0
垫片	Reg_S0_48[5]	1'b0
包装出货	Reg_S0_49[1]	X
同步酶的粘附	Reg_S0_49[2]	1'b0
包块输出结束信号	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
选择ADC同步	Reg_S1_28[2]	1'b0
if_sel_656	Reg_S1_00[3]	注*
if_sel16bit	Reg_S1_00[4]	1'b0
if_sel24bit	Reg_S1_01[7]	1'b0

注："X"表示"0"或"1"均可。注*：若输入656，则设为1；
若输入为601，则将其设为0。

输入引脚	描述	输出引脚	描述
VG[7:0]	数字 YUV 数据输入	AGPb	模拟蓝/铅输出
PCLKIN	时钟输入	AGY	模拟绿色/黄色输出
HSIN1	视频H同步输入	AGPR	模拟红/Pr输出
VSIN1	视频同步输入	ASVM	模拟 SVM 输出
		HSOUT	视频H同步输出
		VSOUT	视频同步输出
		HBOUT	视频H空白输出
		VBOUT	视频空白输出

System Solution 4: Analog YUV/RGB Input with Analog Output Mode



Input Pin	Description	Output Pin	Description
R0/R1/R2	Analog R/V input of CHN0/1/2	AGPb	Analog Blue/Pb output
G0/G1/G2	Analog G/Y input of CHN0/1/2	AGY	Analog Green/Y output
B0/B1/B2	Analog B/U input of CHN0/1/2	AGPr	Analog Red/Pr output
SOG0/SOG1	Analog SOG/Y input of CHN0/1	ASVM	Analog SVM output
HSIN1/HSIN2	Video H-sync input 1/2*	HSOUT	Video H-sync output
VSIN1/VSIN2	Video V-sync input 1/2*	VSOUT	Video V-sync output
		HBOUT	Video H-blank output
		VBOUT	Video V-blank output

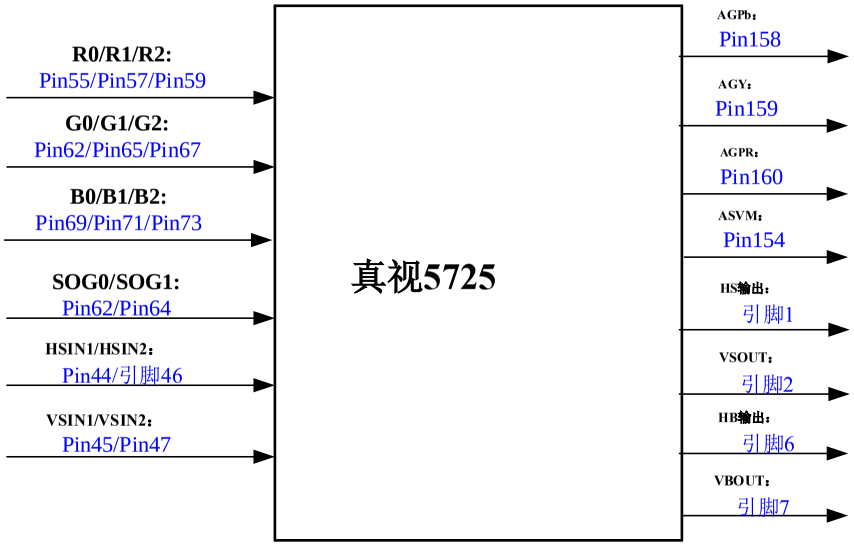
Note*: HSIN1/VSIN1 and HSIN2/VSIN2 could be programmed for analog channel 0/1/2

Figure 10: Analog YUV/RGB Input with Analog Output Mode

5725 Key Register Setting		
Register Name	**Address**	**Value**
pad_bout_en	Reg_S0_48[0]	X
pad_bin_enz	Reg_S0_48[1]	X
pad_rout_en	Reg_S0_48[2]	X
pad_rin_enz	Reg_S0_48[3]	X
pad_gout_en	Reg_S0_48[4]	X
pad_gin_enz	Reg_S0_48[5]	X
pad_ckout_enz	Reg_S0_49[1]	X
pad_sync_out_enz	Reg_S0_49[2]	1'b0
pad_blk_out_enz	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
if_sel_adc_sync	Reg_S1_28[2]	1'b1
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	X
if_sel24bit	Reg_S1_01[7]	1'b1

Note: "X" means either "0" or "1" is OK.

系统解决方案4：模拟 YUV /RGB输入与模拟输出模式



输入引脚	描述	输出引脚	描述
R0/R1/R2	CHN0/1/2模拟R/V输入	AGPb	模拟蓝/铅输出
G0/G1/G2	CHN0/1/2模拟G/Y输入	AGY	模拟绿色/黄色输出
B0/B1/B2	CHN0/1/2模拟B/U输入	AGPR	模拟红/Pr输出
SOG0/SOG1	CHN0/1模拟SOG/Y输入	ASVM	模拟 SVM 输出
HSIN1/HSIN2	视频H同步输入1/2*	HSOUT	视频H同步输出
VSIN1/VSIN2	视频同步输入1/2*	VSOUT	视频同步输出
		HBOUT	视频H空白输出
		VBOUT	视频空白输出

注*：HSIN1/VSIN1与HSIN2/VSIN2可编程为模拟通道0/1/2

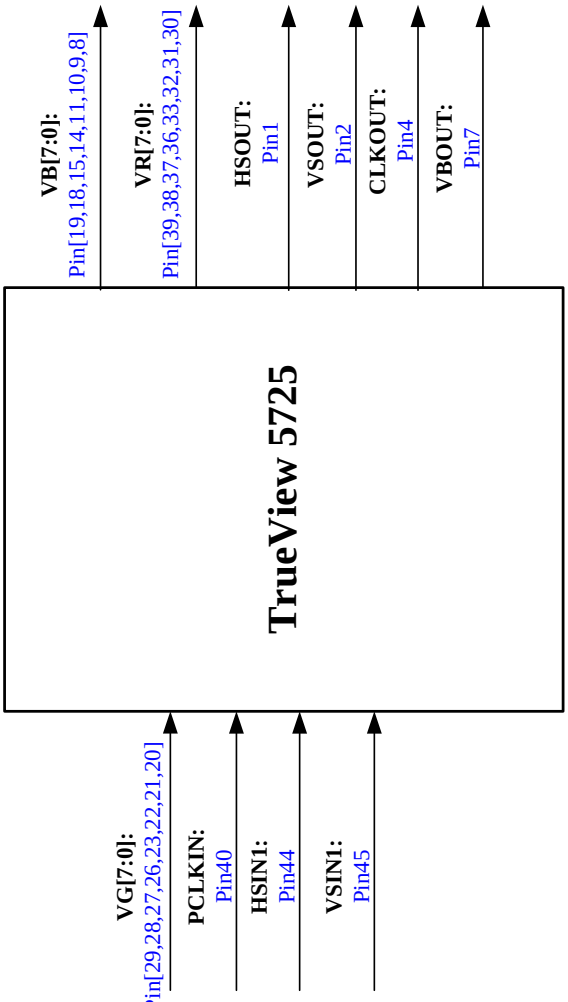
图10：模拟 YUV /RGB输入与模拟输出模式

5725 密钥寄存器设置		
注册名称	地址	值
垫脚	Reg_S0_48[0]	X
酶的截断	Reg_S0_48[1]	X
并行输出通道	Reg_S0_48[2]	X
垫片	Reg_S0_48[3]	X
垫式关节炎	Reg_S0_48[4]	X
垫片	Reg_S0_48[5]	X
包装出货	Reg_S0_49[1]	X
同步酶的粘附	Reg_S0_49[2]	1'b0
包块输出结束信号	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b0
选择ADC同步	Reg_S1_28[2]	1'b1
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	X
if_sel24bit	Reg_S1_01[7]	1'b1

注：“X”表示“0”或“1”均可。

Figure 11: Digital 8-bit 601/656 4:2:2 YUV Input with 16bit Digital Output

System Solution 5: Digital 8-bit 601/656 4:2:2 YUV Input with 16bit Digital Output Mode



5725 Key Register Setting		
Register Name	**Address**	**Value**
pad_bout_en	Reg_S0_48[0]	1'b1
pad_bin_enz	Reg_S0_48[1]	1'b0
pad_rout_en	Reg_S0_48[2]	1'b1
pad_rin_enz	Reg_S0_48[3]	1'b1
pad_gout_en	Reg_S0_48[4]	1'b1
pad_gin_enz	Reg_S0_48[5]	1'b0
pad_ckout_enz	Reg_S0_49[1]	1'b0
pad_sync_out_enz	Reg_S0_49[2]	1'b0
pad_blk_out_enz	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b0
out_blank_sel_0	Reg_S0_50[0]	1'b1
if_sel_adc_sync	Reg_S1_28[2]	1'b1
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	1'b1
if_sel24bit	Reg_S1_01[7]	1'b0

Note: "X" means either "0" or "1" is OK.

Input Pin	Description	Output Pin	Description
VG[7:0]	Digital YUV data input	VB[7:0]	Digtial UV data output
PCLKIN	Pixel clock input	VR[7:0]	Digital Y data output
HSIN1	Video H-sync input	HSOUT	Video H-sync output
VSIN1	Video V-sync input	VSOUT	Video V-sync output
		CLKOUT	Video display clock output
		VBOUT	Display enable output for LCD

系统解决方案5：数字8位601/656 4:2:2 YUV 输入，支持16位数字输出模式

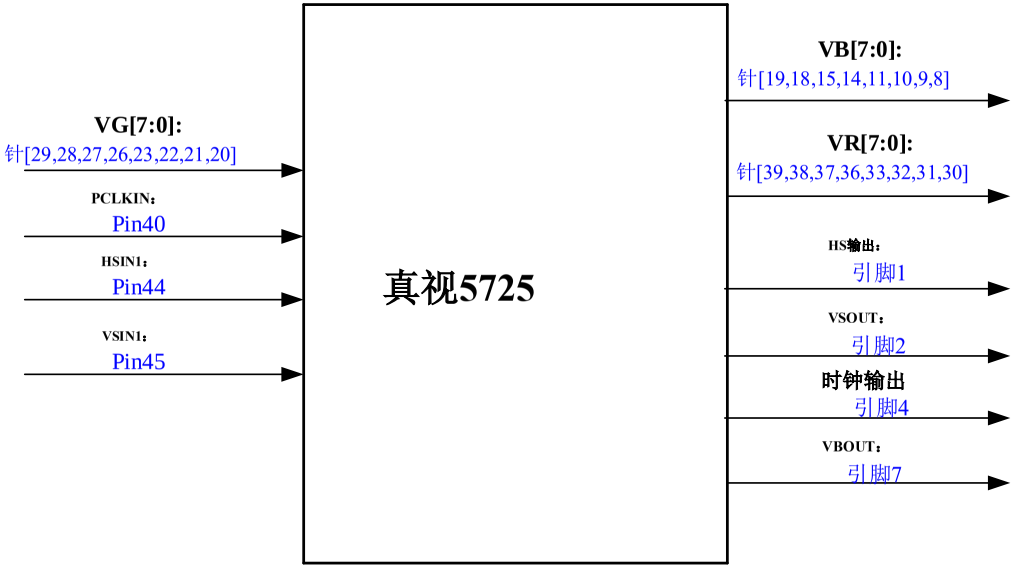


图11：数字输入（8位/601/656,4:2:2 YUV，16位数字输出）

5725 密钥寄存器设置		
注册名称	地址	值
垫脚	Reg_S0_48[0]	1'b1
酶的截断	Reg_S0_48[1]	1'b1
并行输出通道	Reg_S0_48[2]	1'b1
垫片	Reg_S0_48[3]	1'b1
垫式关节炎	Reg_S0_48[4]	1'b0
垫片	Reg_S0_48[5]	1'b0
包装出货	Reg_S0_49[1]	1'b0
同步酶的粘附	Reg_S0_49[2]	1'b0
包块输出结束信号	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b1
out_blank_sel_0	Reg_S0_50[0]	1'b1
选择ADC同步	Reg_S1_28[2]	1'b0
if_sel_656	Reg_S1_00[3]	1'b1
if_sel16bit	Reg_S1_00[4]	1'b0
if_sel24bit	Reg_S1_01[7]	1'b0

注：“X”表示“0”或“1”均可。

输入引脚	描述	输出引脚	描述
VG[7:0]	数字 YUV 数据输入	VB[7:0]	数字紫外数据输出
PCLKIN	时钟输入	VR[7:0]	数字Y数据输出
HSIN1	视频H同步输入	HSOUT	视频H同步输出
VSIN1	视频同步输入	VSOUT	视频同步输出
		时钟输出	视频显示时钟输出
		VBOUT	显示LCD的输出

System Solution 6: Analog RGB/YUV Input with 16bit Digital Output Mode

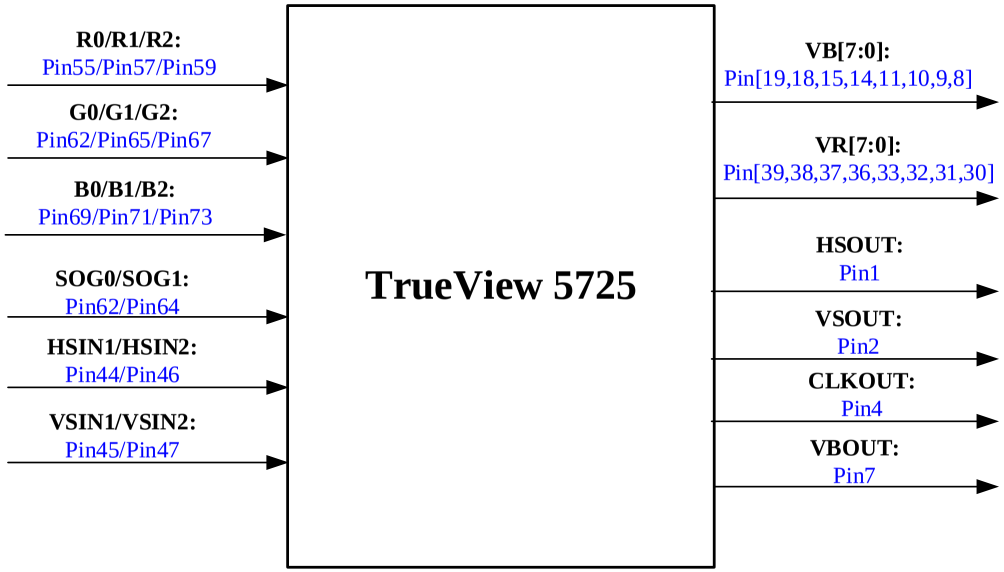


Figure 12: Analog RGB/YUV Input with 16bit Digital Output Mode

5725 Key Register Setting		
Register Name	**Address**	**Value**
pad_bout_en	Reg_S0_48[0]	1'b1
pad_bin_enz	Reg_S0_48[1]	1'b1
pad_rout_en	Reg_S0_48[2]	1'b1
pad_rin_enz	Reg_S0_48[3]	1'b1
pad_gout_en	Reg_S0_48[4]	X
pad_gin_enz	Reg_S0_48[5]	X
pad_ckout_enz	Reg_S0_49[1]	1'b0
pad_sync_out_enz	Reg_S0_49[2]	1'b0
pad_blk_out_enz	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b1
out_blank_sel_0	Reg_S0_50[0]	1'b1
if_sel_adc_sync	Reg_S1_28[2]	1'b1
if_sel_656	Reg_S1_00[3]	X
if_sel16bit	Reg_S1_00[4]	X
if_sel24bit	Reg_S1_01[7]	1'b1

Note: "X" means either "0" or "1" is OK.

Input Pin	Description	Output Pin	Description
R0/R1/R2	Analog R/V input of CHN0/1/2	VB[7:0]	Digtial UV data output
G0/G1/G2	Analog G/Y input of CHN0/1/2	VR[7:0]	Digital Y data output
B0/B1/B2	Analog B/U input of CHN0/1/2	HSOUT	Video H-sync output
SOG0/SOG1	Analog SOG/Y input of CHN1/2	VSOUT	Video V-sync output
HSIN1/HSIN2	Video H-sync input 1/2*	CLKOUT	Video display clock output
VSIN1/VSIN2	Video V-sync input 1/2*	VBOUT	Display enable output for LCD

Note*: HSIN1/VSIN1 and HSIN2/VSIN2 could be programmed for analog channel 0/1/2

系统解决方案6：模拟RGB/ YUV 输入，支持16位数字输出模式

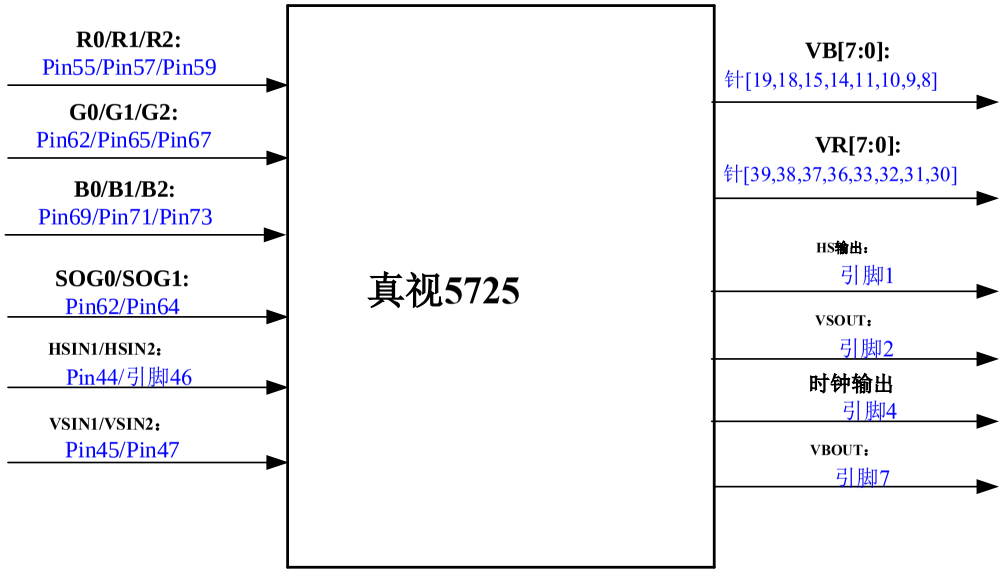


图12：16位数字输出模式的RGB/ YUV 输入模拟信号

5725 密钥寄存器设置		
注册名称	地址	值
垫脚	Reg_S0_48[0]	1'b1
酶的截断	Reg_S0_48[1]	1'b1
并行输出通道	Reg_S0_48[2]	1'b1
垫片	Reg_S0_48[3]	1'b1
垫式关节炎	Reg_S0_48[4]	X
垫片	Reg_S0_48[5]	X
包装出货	Reg_S0_49[1]	1'b0
同步酶的粘附	Reg_S0_49[2]	1'b0
包块输出结束信号	Reg_S0_49[3]	1'b0
vds_do_16b_en	Reg_S3_50[7]	1'b1
out_blank_sel_0	Reg_S0_50[0]	1'b1
选择ADC同步	Reg_S1_28[2]	1'b1
if_sel_656	Reg_S1_00[3]	X
if_sel16bit	Reg_S1_00[4]	X
if_sel24bit	Reg_S1_01[7]	1'b1

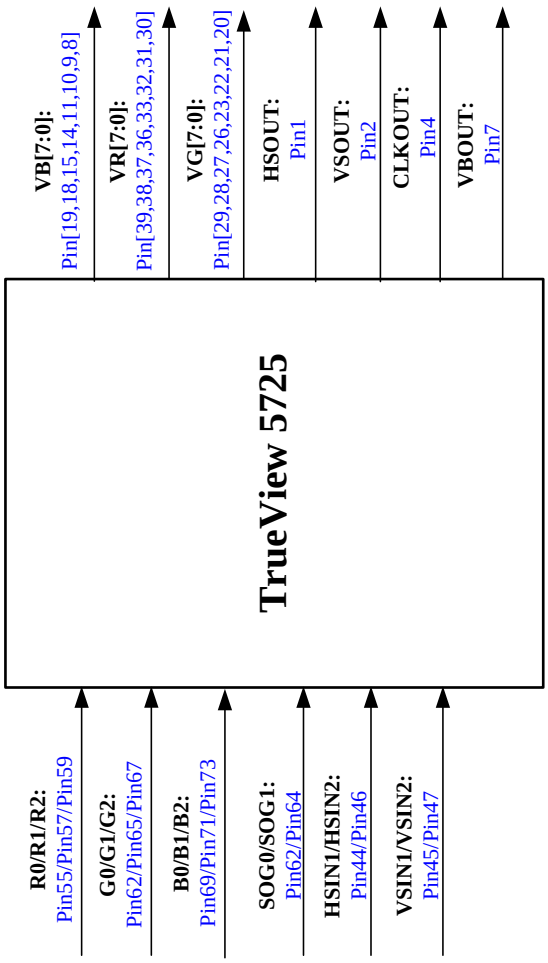
注：“X”表示“0”或“1”均可。

输入引脚	描述	输出引脚	描述
R0/R1/R2	CHN0/1/2模拟R/V输入	VB[7:0]	数字紫外数据输出
G0/G1/G2	CHN0/1/2模拟G/Y输入	VR[7:0]	数字Y数据输出
B0/B1/B2	CHN0/1/2模拟B/U输入	HSOUT	视频H同步输出
SOG0/SOG1	CHN1/2模拟SOG/Y输入	VSOUT	视频同步输出
HSIN1/HSIN2	视频H同步输入1/2*	时钟输出	视频显示时钟输出
VSIN1/VSIN2	视频同步输入1/2*	VBOUT	显示LCD的输出

注*：HSIN1/VSIN1与HSIN2/VSIN2可编程为模拟通道0/1/2

Figure 13: Analog RGB/YUV Input with 24bit Digital Output Mode

System Solution 7: Analog RGB/YUV Input with 24bit Digital Output Mode



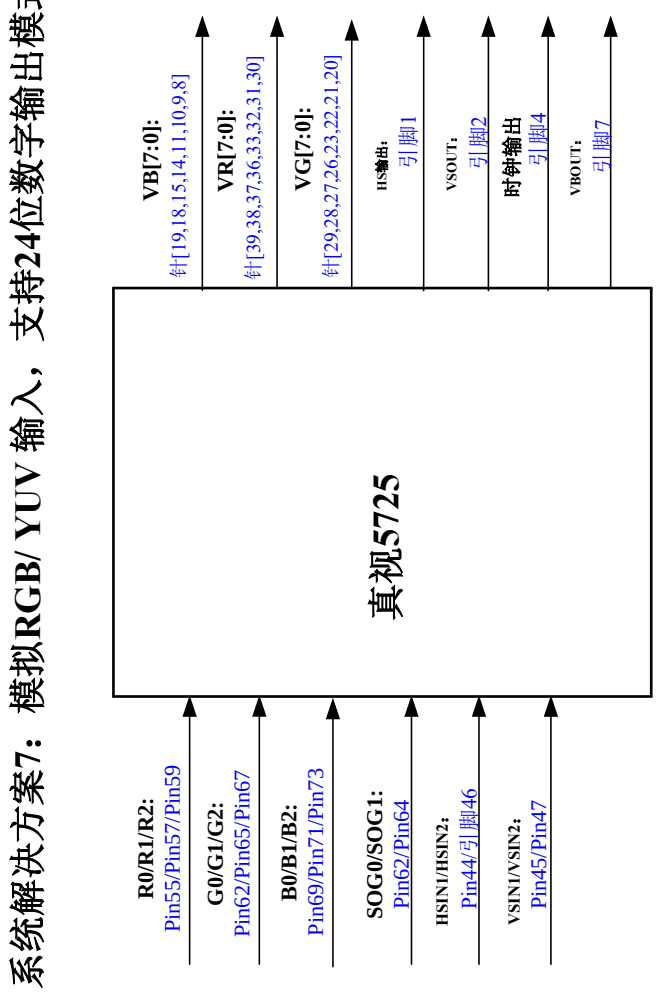
5725 Key Register Setting		
Register Name	**Address**	**Value**
pad_bout_en	Reg_S0_48[0]	1'b1
pad_bin_enz	Reg_S0_48[1]	1'b 1
pad_rout_en	Reg_S0_48[2]	1'b 1
pad_rin_enz	Reg_S0_48[3]	1'b 1
pad_gout_en	Reg_S0_48[4]	1'b 1
pad_gin_enz	Reg_S0_48[5]	1'b 1
pad_ckout_enz	Reg_S0_49[1]	1'b 0
pad_sync_out_enz	Reg_S0_49[2]	1'b 0
pad_blk_out_enz	Reg_S0_49[3]	1'b 0
vds_do_16b_en	Reg_S3_50[7]	1'b 0
out_blank_sel_0	Reg_S0_50[0]	1'b 1
if_sel_adc_sync	Reg_S1_28[2]	1'b 1
if_sel_656	Reg_S1_00[3]	1'b 0
if_sel16bit	Reg_S1_00[4]	X
if_sel24bit	Reg_S1_01[7]	1'b 1

Note: "X" means either "0" or "1" is OK.

Input Pin	Description	Output Pin	Description
R0/R1/R2	Analog R/V input of CHN0/1/2	VB[7:0]	Digtial R/V data output
G0/G1/G2	Analog G/Y input of CHN0/1/2	VG[7:0]	Digtial G/Y data output
B0/B1/B2	Analog B/U input of CHN0/1/2	VR[7:0]	Digital B/U data output
SOG0/SOG1	Analog SOG/Y input of CHN1/2	HSOUT	Video H-sync output
HSIN1/HSIN2	Video H-sync input 1/2*	VSOUT	Video V-sync output
VSIN1/VSIN2	Video V-sync input 1/2*	CLKOUT	Video display clock output
		VBOUT	Display enable output for LCD

Note*: HSIN1/VSIN1 and HSIN2/VSIN2 could be programmed for analog channel 0/1/2

图13：24位数字输出模式下的RGB/ YUV 输入分析



5725 密钥寄存器设置		
注册名称	地址	值
垫脚	Reg_S0_48[0]	1'b1
酶的截断	Reg_S0_48[1]	1'b
并行输出通道	Reg_S0_48[2]	1'b 1'b
垫片	Reg_S0_48[3]	
垫式关节炎	Reg_S0_48[4]	1'b 1'b
垫片	Reg_S0_48[5]	1'b 1'b
pad_ckout_enz、pad_	Reg_S0_49[1]	
sync_out_enz、pad	Reg_S0_49[2]	1'b 1'b
blk_out_enz vds_do_	Reg_S0_49[3]	
16b_en	Reg_S3_50[7]	0
out_blank_sel_0	Reg_S0_50[0]	1 b 1
选择ADC同步	Reg_S1_28[2]	1'b 1
if_sel_656	Reg_S1_00[3]	1'b0
if_sel16bit	Reg_S1_00[4]	x
if_sel24bit	Reg_S1_01[7]	1'b 1

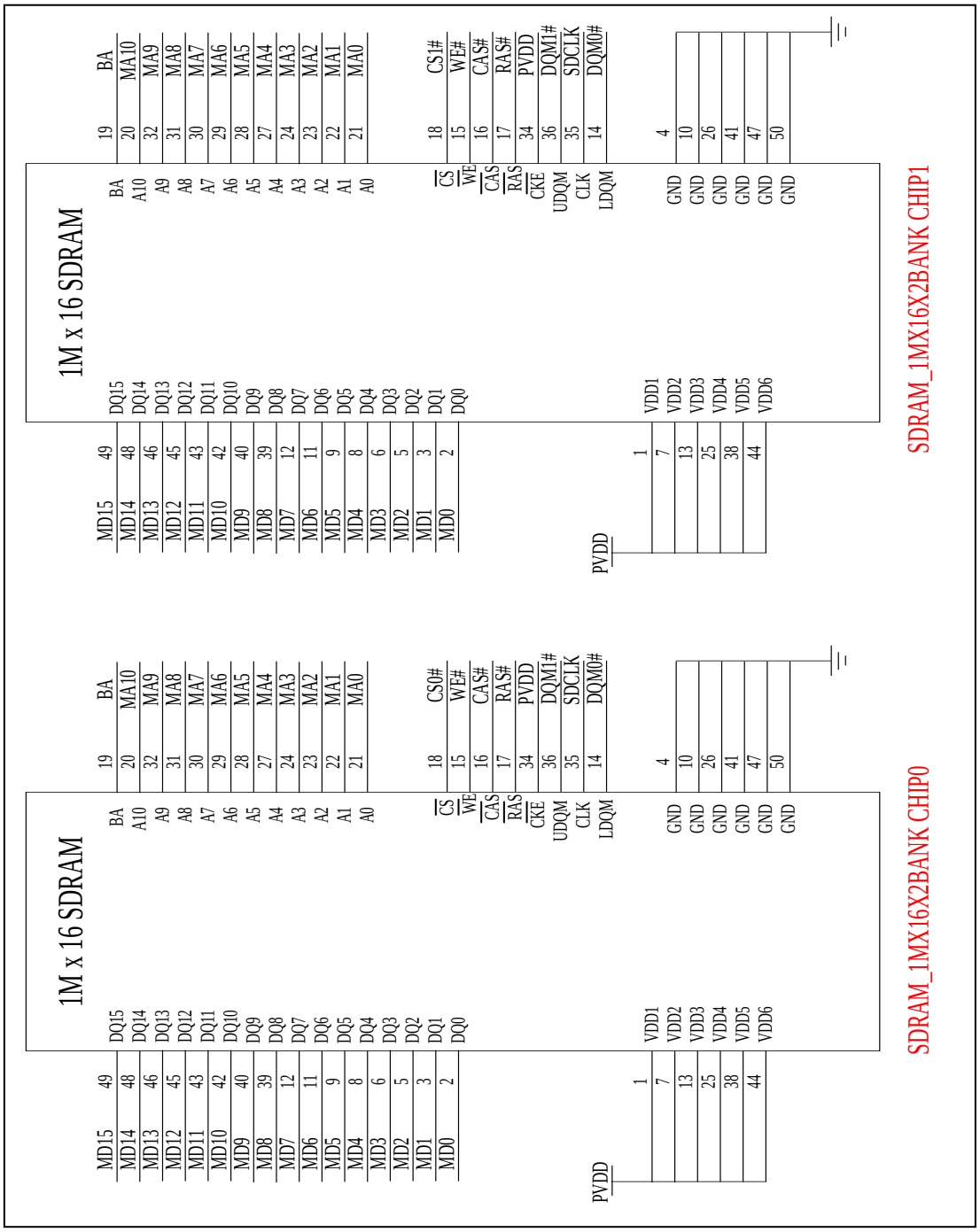
注："X" 表示 "0" 或 "1" 均可。

输入引脚	描述	输出引脚	描述
R0/R1/R2	CHN0/1/2模拟R/V输入	VB[7:0]	数字式无线电测深仪数据输出
G0/G1/G2	CHN0/1/2模拟G/Y输入	VG[7:0]	数字G/Y数据输出
B0/B1/B2	CHN0/1/2模拟B/U输入	VR[7:0]	数字黑白数据输出
SOG0/SOG1	CHN1/2模拟SOG/Y输入	HSOUT	视频H同步输出
HSIN1/HSIN2	视频H同步输入1/2*	VSOUT	视频同步输出
VSIN1/VSIN2	视频同步输入1/2*	时钟输出	视频显示时钟输出
		VBOUT	显示LCD的输出

注*：HSIN1/VSIN1与HSIN2/VSIN2可编程为模拟通道0/1/2

BOARD MEMORY CONNECTION:

Figure 14: THREE 1MX16X2BANK MEMORY (1)



板存储器连接

图14：三块1MX16X2BANK存储器(1)

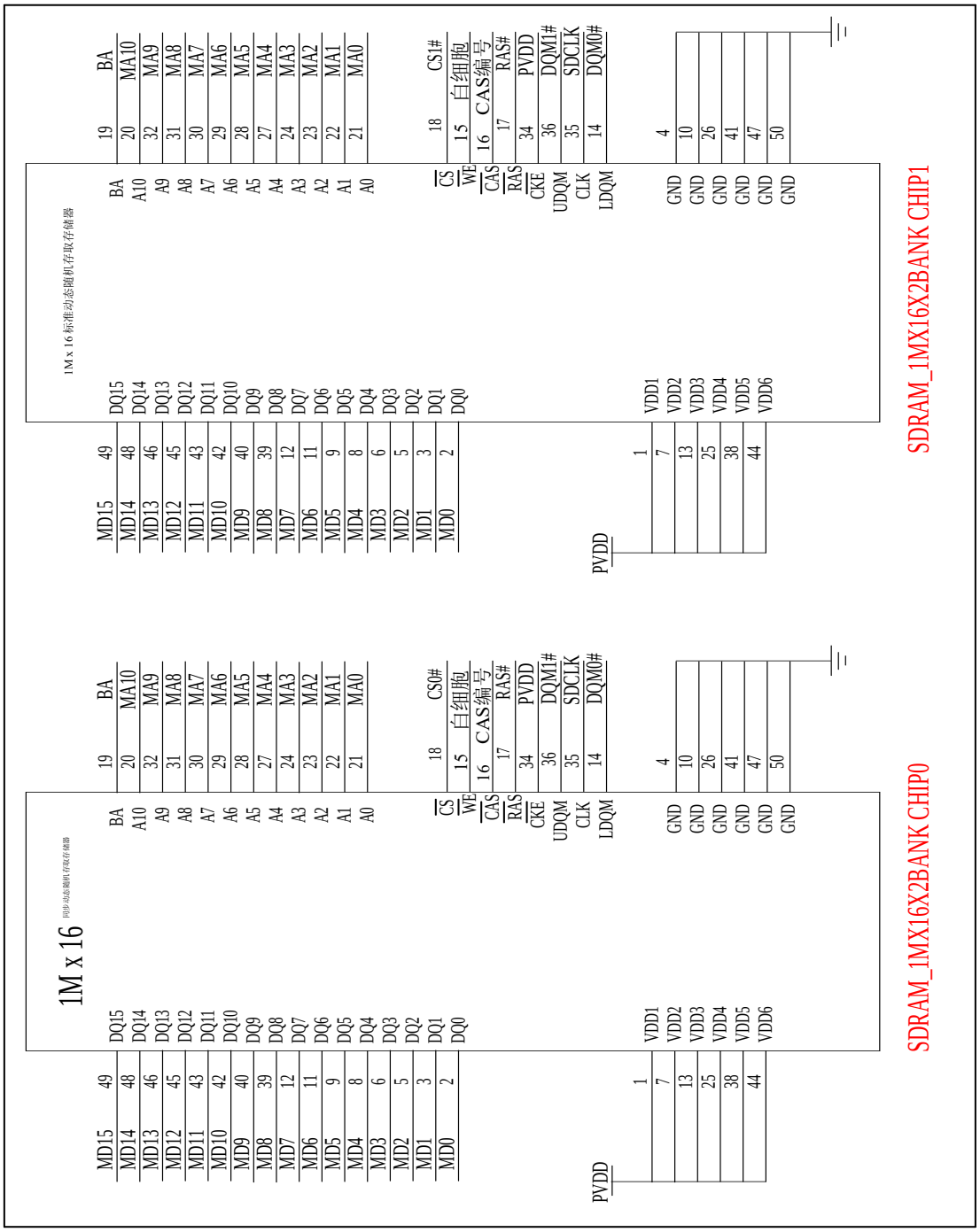


Figure 15: THREE 1MX16X2BANK MEMORY (2)

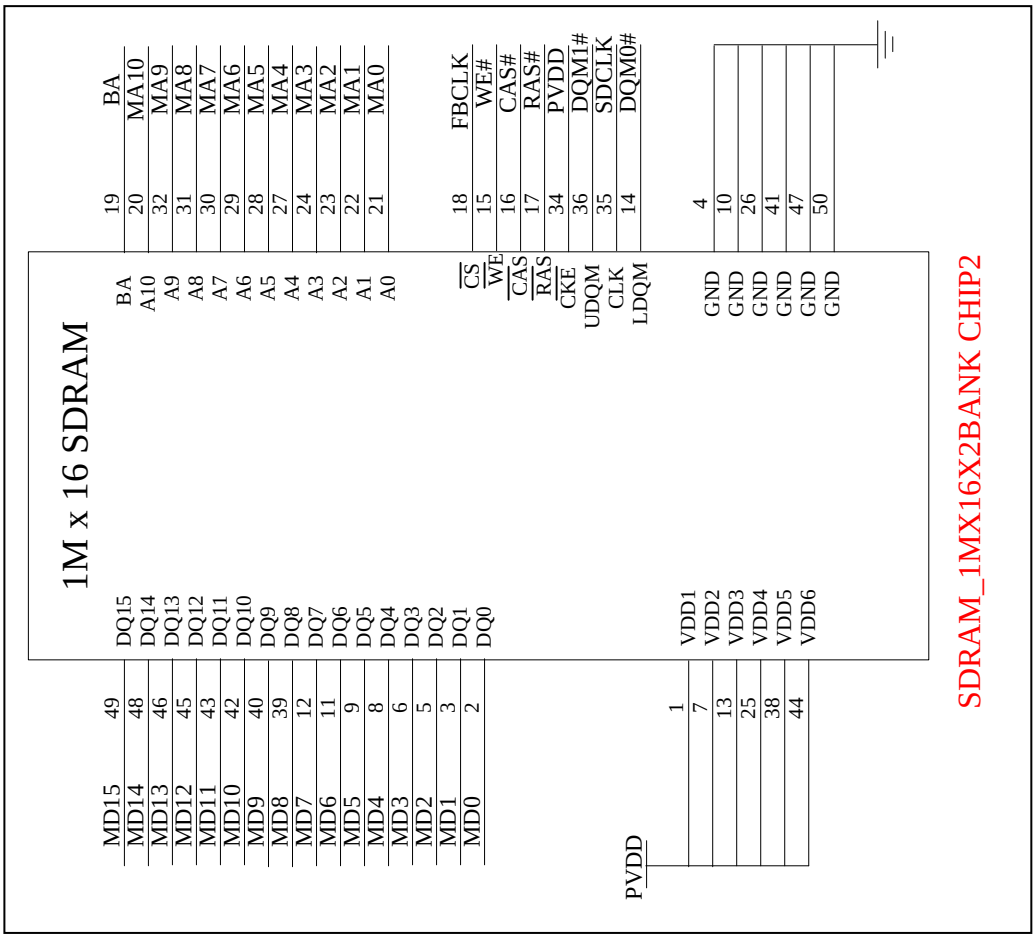


图15：三块1MX16X2BANK存储器(2)

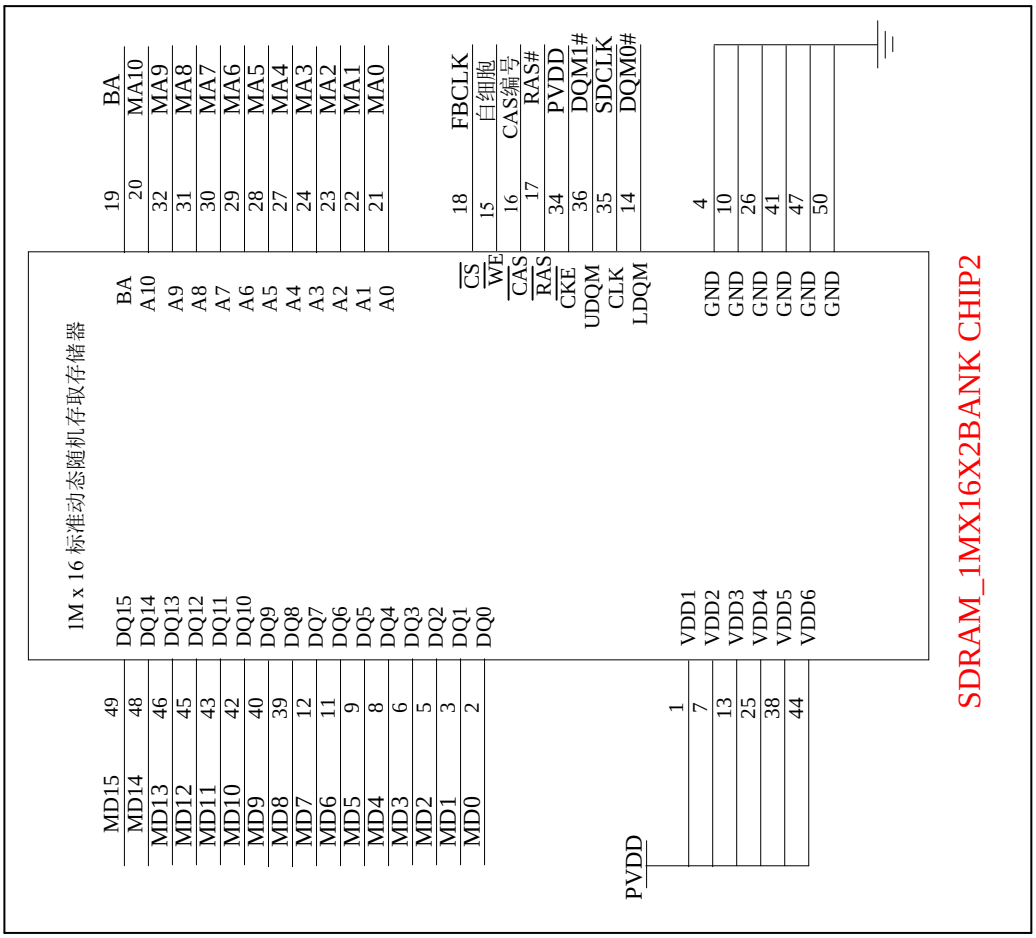


Figure 16: ONE 1MX16X4BANK MEMORY

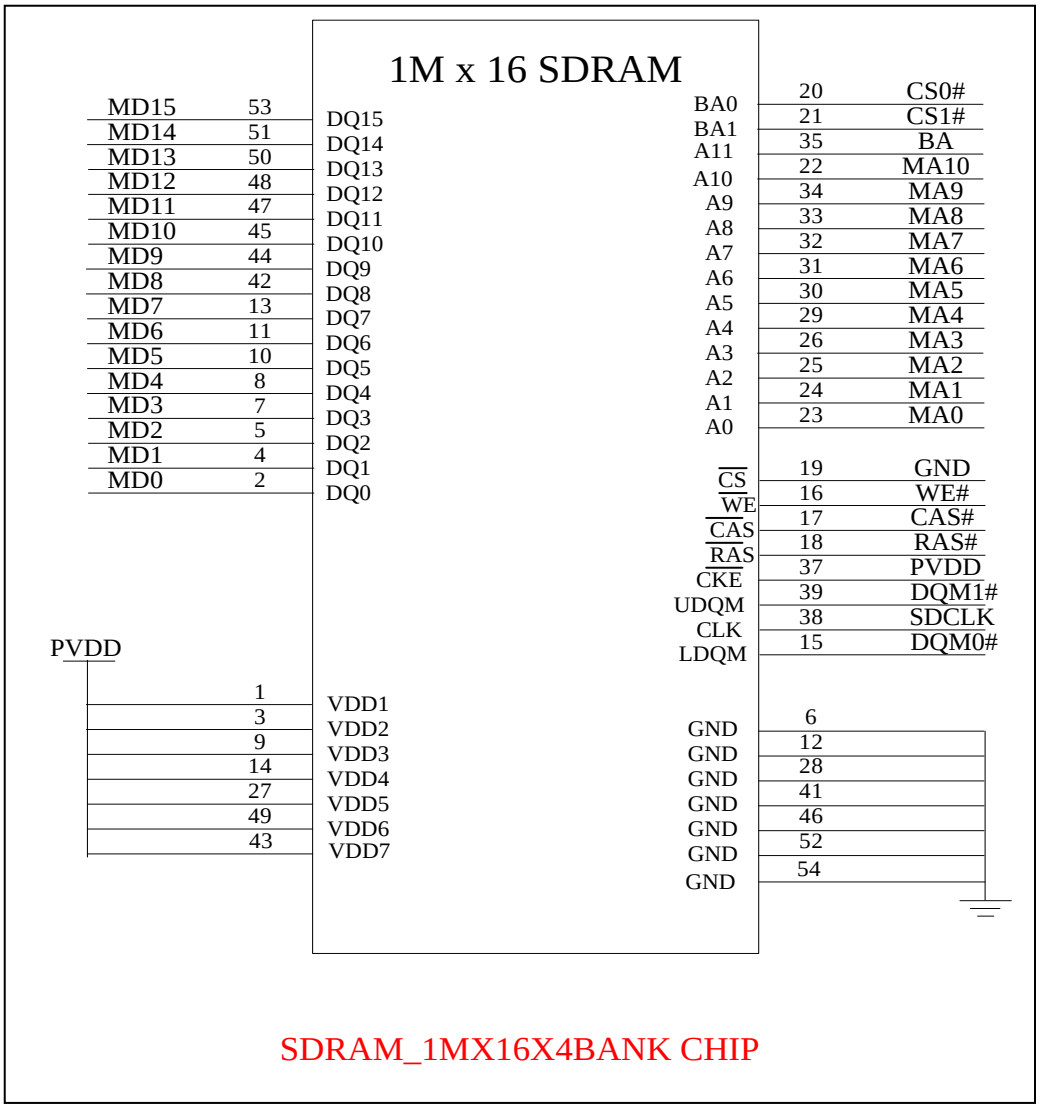


图16：ONE 1MX16X4BANK存储器

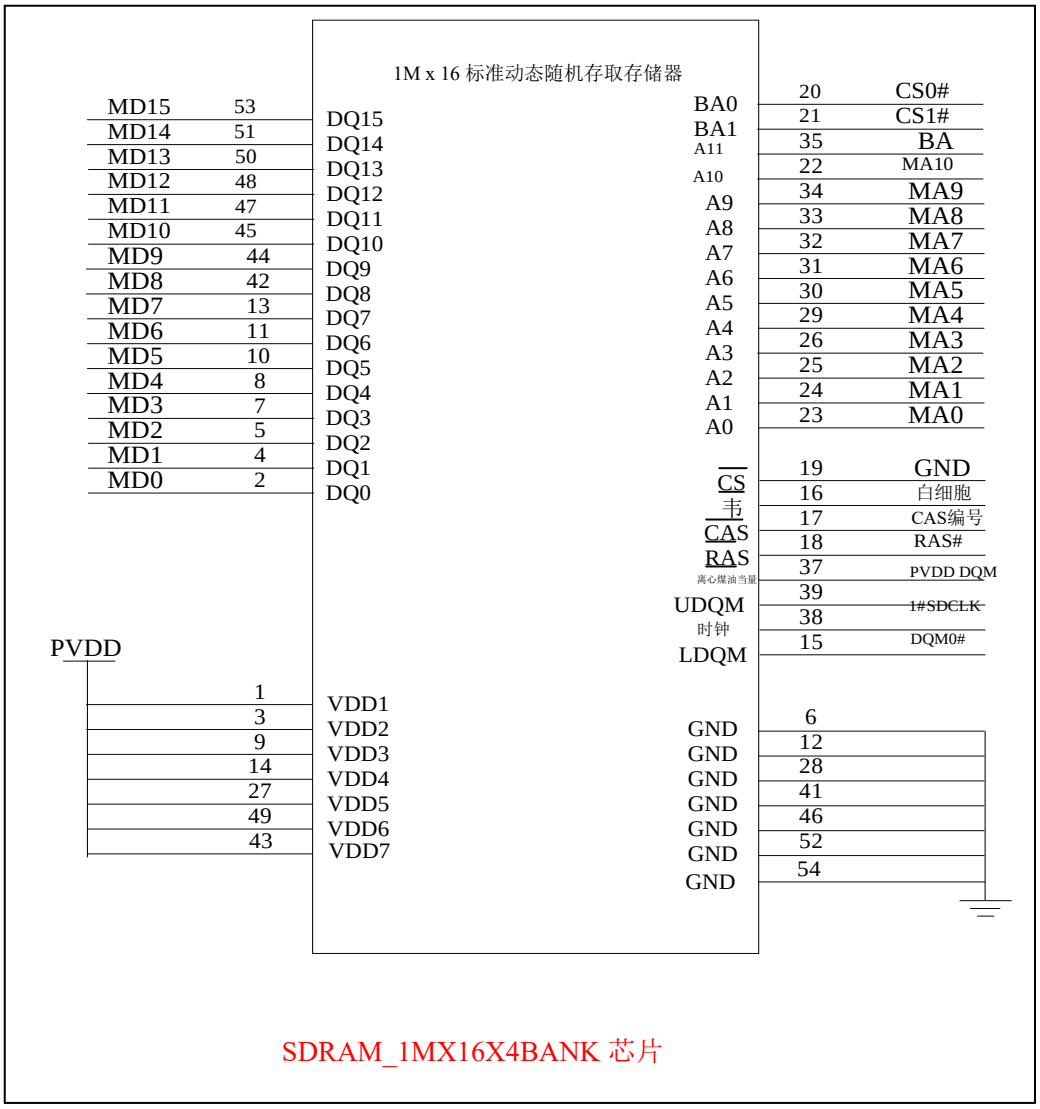


Figure 17: ONE 512KX32X4BANK MEMORY

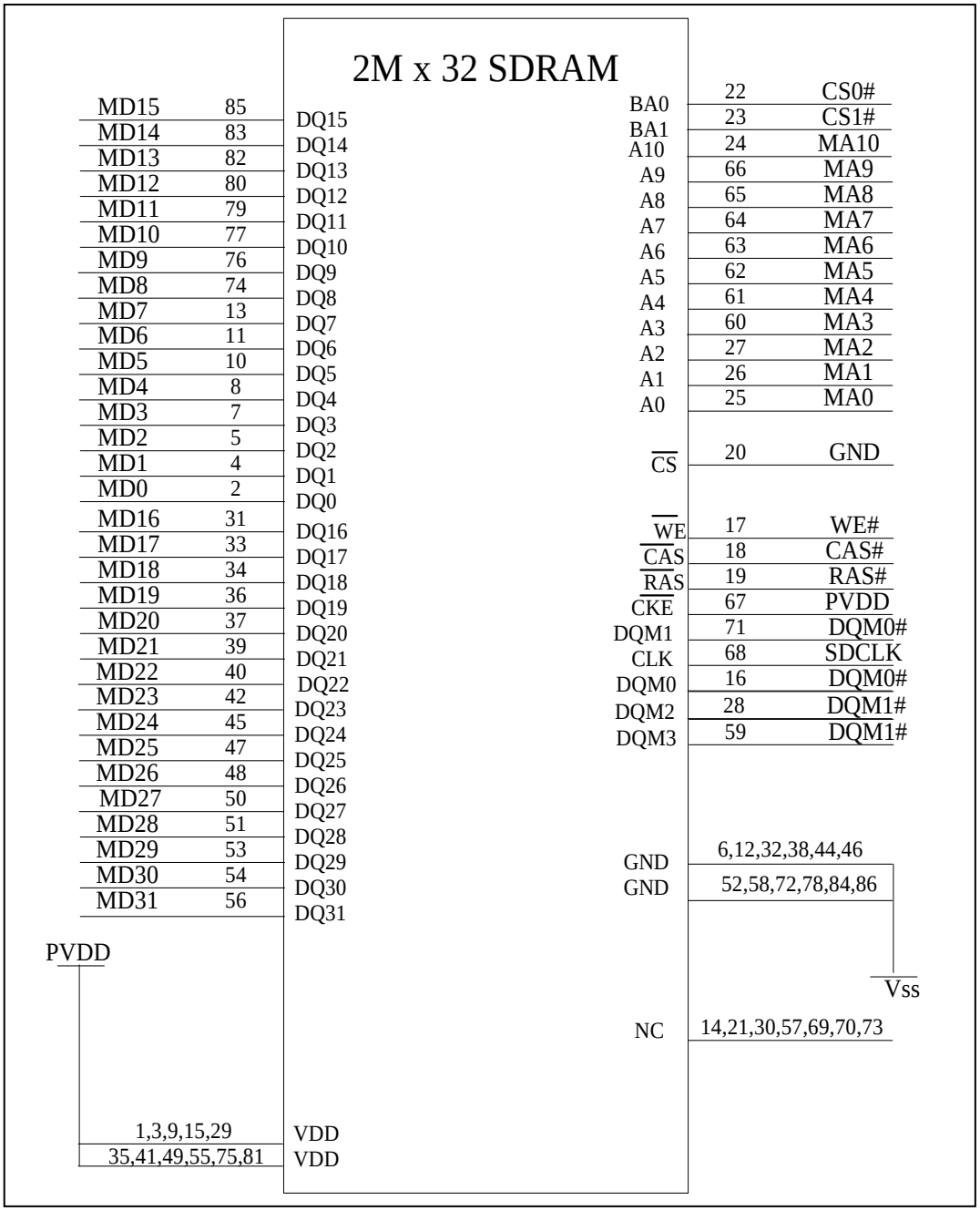


图17：ONE 512KX32X4BANK存储器

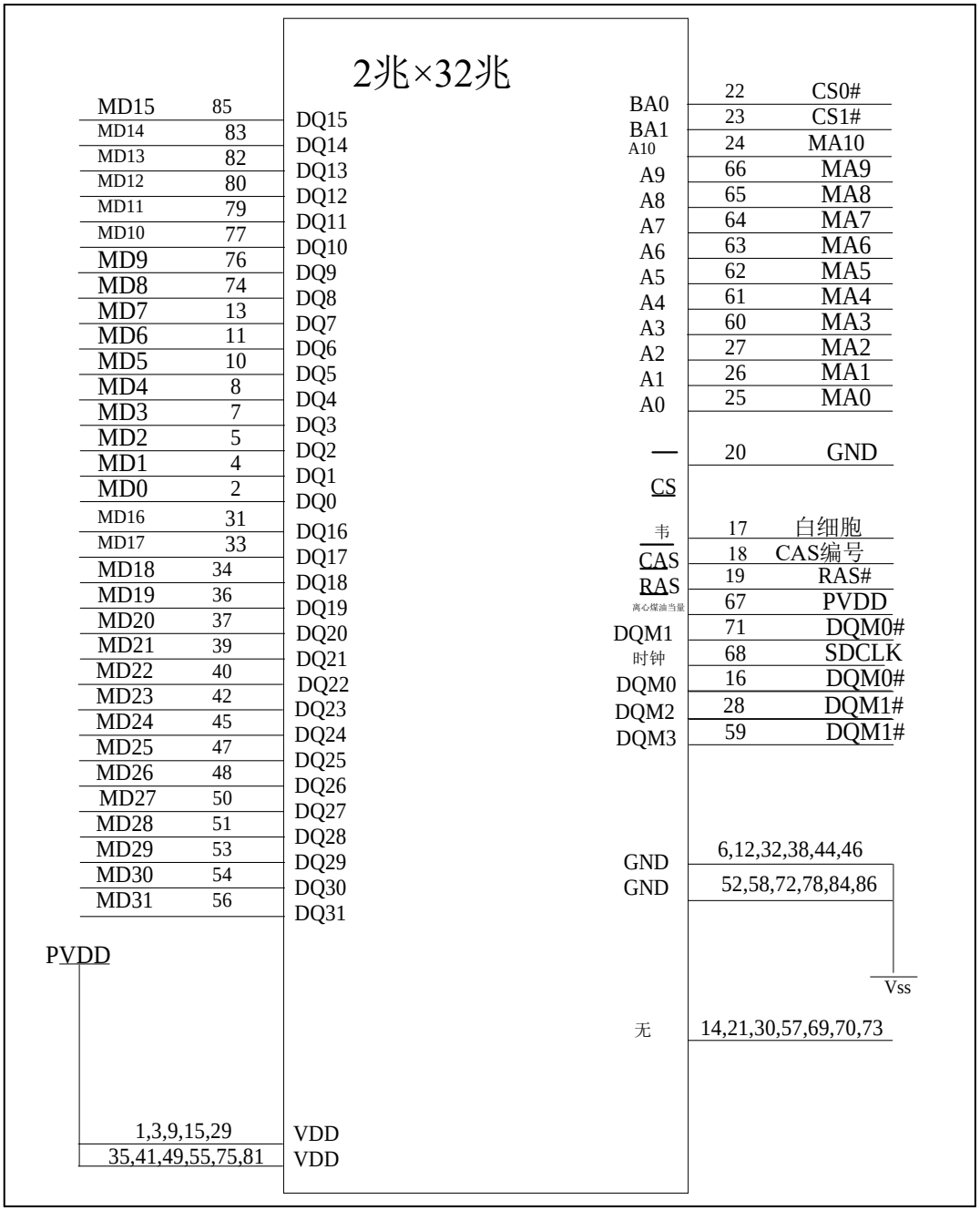


Figure 18: TWO 1MX16X4BANK MEMORY

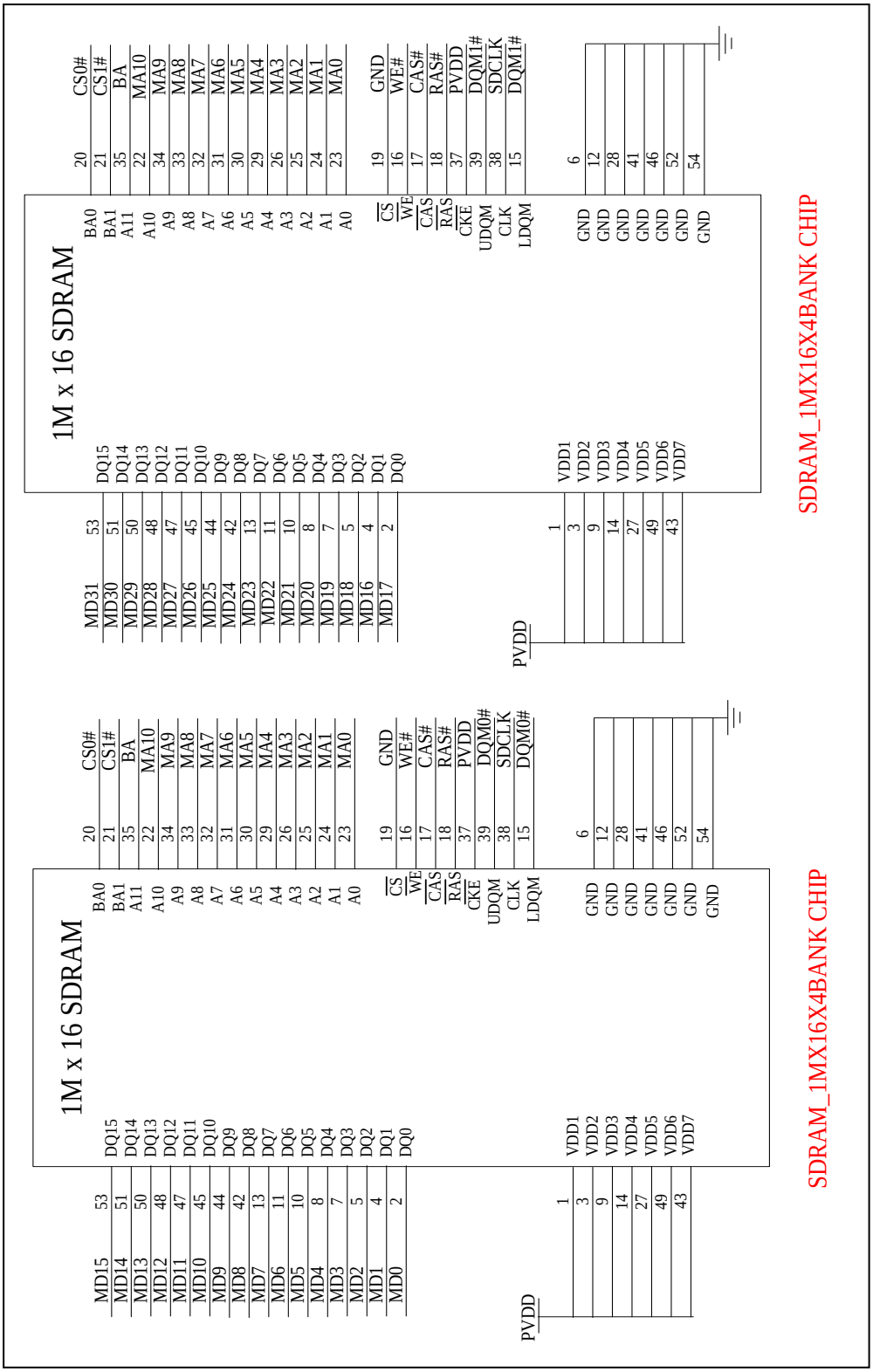
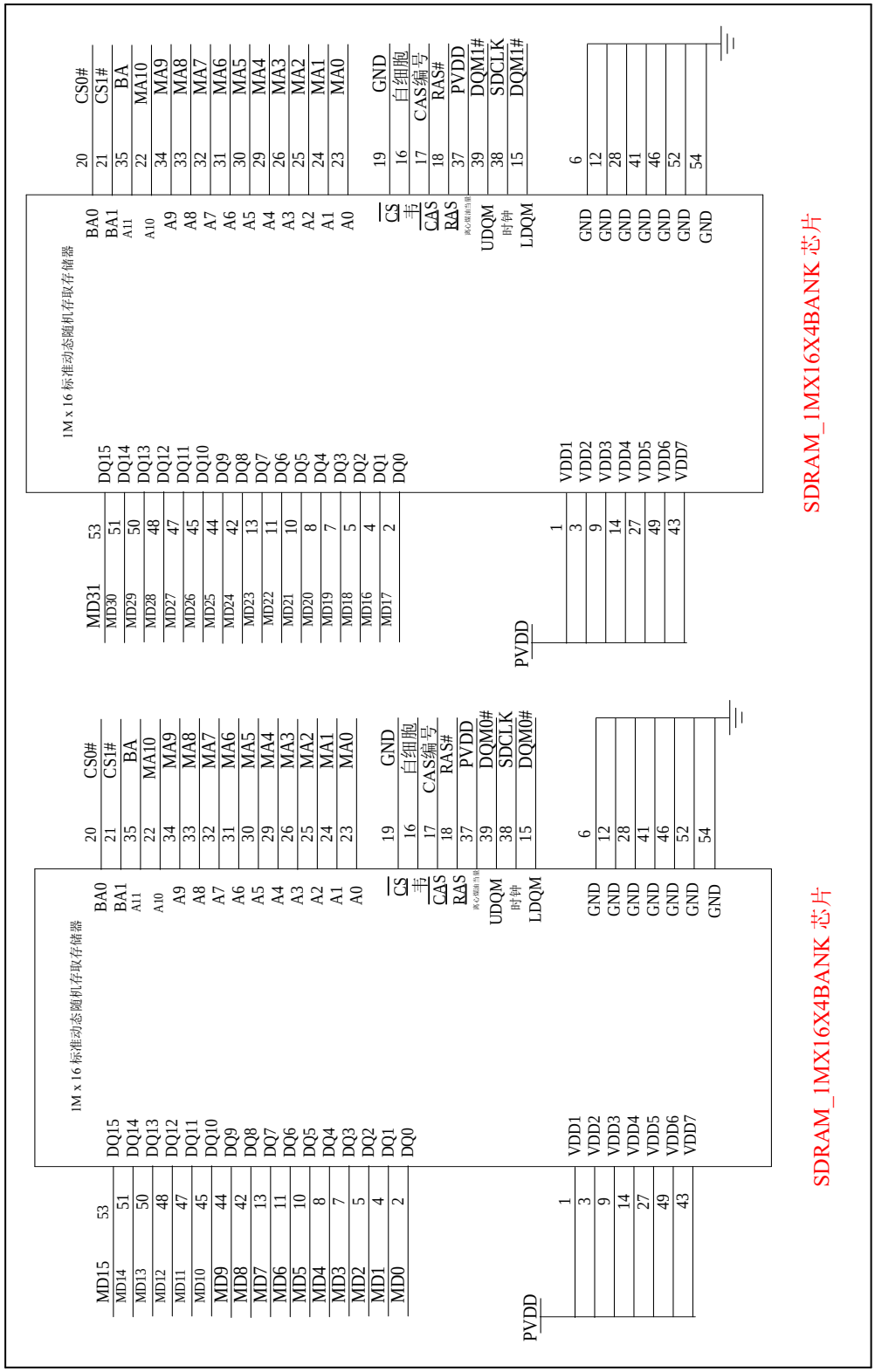


图18：两块1MX16X4BANK存储器



ELECTRICAL CHARACTERISTICS

ABSOLUTE MAXIMUM RATINGS

Table 12: Absolute Maximum Ratings

Parameter	Symbol	Min	Max	Units	Conditions
3.3V Power Supply Voltage (Reference to Ground)	VDD[1]	- 0.3	4.0	V	
1.8V Power Supply Voltage (Reference to Ground)	VCC	- 0.3	2.2	V	
Voltage on any input	V_I	- 0.3	VDD+0.3	V	
Storage Temperature	T_S	- 40	125	℃	
Operating temperature	T_o	0	70	℃	

Warning: Stressing the device beyond the "Absolute maximum Ratings" may cause permanent damage.

Notes

1. *TV5725's 3.3V power supply : PVDD, DAVD, PAVD, AVD_PLL, AVD_REF, AVD_R, AVD_G, AVD_B.*

电气特性

A 绝对 M AXIMUM R ATINGS

表12：绝对最大额定值

参数	符号	微小的	最高的	单位	条件
3.3V电源电压（参考地）	VDD[1]	- 0.3	4.0	V	
1.8V电源电压（参考地）	小型盒式录像带	- 0.3	2.2	V	
任何输入端的电压	V_I	- 0.3	VDD+0.3	V	
储存温度	T_S	- 40	125	℃	
工作温度	T_o	0	70	℃	

警告：若将设备应力超过“绝对最大额定值”，可能导致永久性损坏。

记下

1. *TV5725,3.3V电源：PVDD、DAVD、PAVD、AVD_PLL、AVD_REF、AVD_R、AVD_G、AVD_B。*

DC CHARACTERISTICS

Table 13: DC Characteristics

Parameter	Symbol	Min	Typ	Max	Units	Comments
1.8V Power Supply Voltage (1.8V±5%)	VCC	1.71		1.89	V	1.8V Operation
3.3V Power Supply Voltage (3.3V±5%)	PVDD, DAVD, PAVD, AVD_PLL, AVD_REF, AVD_R, AVD_G, AVD_B	3.135		3.465	V	3.3V Operation
1.8V Supply Current[1.a] (VCC)	IVCC		320		mA	
3.3V Supply Current[1.a] (PVDD, PAVD, AVD_PLL)	IVDD33		100		mA	
3.3V Supply Current[1.b] (AVD_REF, AVD_R, AVD_G, AVD_B)	IAVD		240		mA	
3.3V Supply Current[1.c] (DAVD)	IDAVD		85		mA	
Input Low Voltage	VIL	0		0.8	V	
Input High Voltage	VIH	2.0		Vcc + 0.3	V	
Schmitt Input Low Voltage[2]	VILS	0		1.0	V	
Schmitt Input High Voltage[2]	VIHS	2.4		Vcc + 0.3	V	
Output Low Voltage	VOL			0.4	V	IOL = 8mA
Output High Voltage	VOH	2.4			V	IOH = -8mA
Input Capacitance	CIN		7		pF	
Output Capacitance	COUT		7		pF	

DC CHARACTERISTICS

表13：直流特性

参数	符号	微小的	类型	最高的	单位	评论
1.8V电源电压（1.8V±5%）	小型盒式录像带	1.71		1.89	V	1.8V操作
3.3V电源电压（3.3V±5%）	PVDD, DAVD, PAVD, AVD_PLL, AVD_REF, AVD_R, AVD_G, AVD_B	3.135		3.465	V	3.3V工作
1.8V 电源电流 [1.a] 小型盒式录像带	血管内消耗性凝血病		320		mA	
3.3V 电源电流 [1.a]（PVDD，PAVD，AVD_PLL)	IVDD33		100		mA	
3.3V 电源电流 [1.b] (AVD_REF，AVD_R，AVD_G，AVD_B)	IAVD		240		mA	
3.3V 电源电流 [1.c] (DAVD)	IDAVD		85		mA	
输入低电压	VIL	0		0.8	V	
输入高电压	暴力致残	2.0		Vcc + 0.3	V	
施密特输入低电压[2]	VILS	0		1.0	V	
施密特输入高压[2]	VIHS	2.4		Vcc + 0.3	V	
输出低电压	卷			0.4	V	IOL = 8mA
输出高电压	验证摘机	2.4			V	IOH = -8mA
输入电容	CIN		7		pF	
输出电容	标准输出		7		pF	

Notes

1. Current measured condition:

a. @ HD 1080i, 74.25MHz input, SOG, 162MHz 32bit memory, 108MHz Display clock, VCC is 1.8V.
b. @ UXGA 60Hz, power supply is 3.3V.
c. For VESA Video Levels, the R_{iref} in the board is 160ohms. The output load is double terminated with 75ohms and 10pf per channel. Full-scale for all four channels is 0.7V, power supply is 3.3V. Measurement temperature is 25^0C. This is a preliminary measurement and is subject to change by Tvia, Inc.

2. Schmitt trigger PADs (Total: 41 Pads):

HSOUT, VSOUT

CLKOUT

HBOUT, VBOUT

VB[0:7]

VG[0:7]

VR[0:7]

PCLKIN

SCLSA

HSIN1, VSIN1, HSIN2, VSIN2

RSTN

GPIO

HALF

MBA

MCS1#

FBCLK

记下

1. 当前测量状态：

a. @HD1080i，74.25MHz输入频率，SOG，162MHz 32位内存，108MHz显示时钟， VCC 为1.8V。
b. @ UXGA 60Hz，电源电压为3.3V。
c. 对于VESA视频电平，电路板中的R_{iref}为160欧姆。输出负载为双端接，每通道75欧姆和10皮法。所有四个通道的满量程为0.7V，电源为3.3V。测量温度为25 ^{0}C。这是初步测量，可能会由Tvia公司更改。

2. 施密特触发器电极（共41个电极）：

HSOUT, VSOUT

时钟输出

HBOUT, VBOUT

VB[0:7]

VG[0:7]

VR[0:7]

PCLKIN

SCLSA

HSIN1, VSIN1, HSIN2, VSIN2

太阳射电望远镜网

通用输入输出

半数

MBA

MCS1

FBCLK

AC CHARACTERISTICS

Figure 19: Video Input Port AC Timing

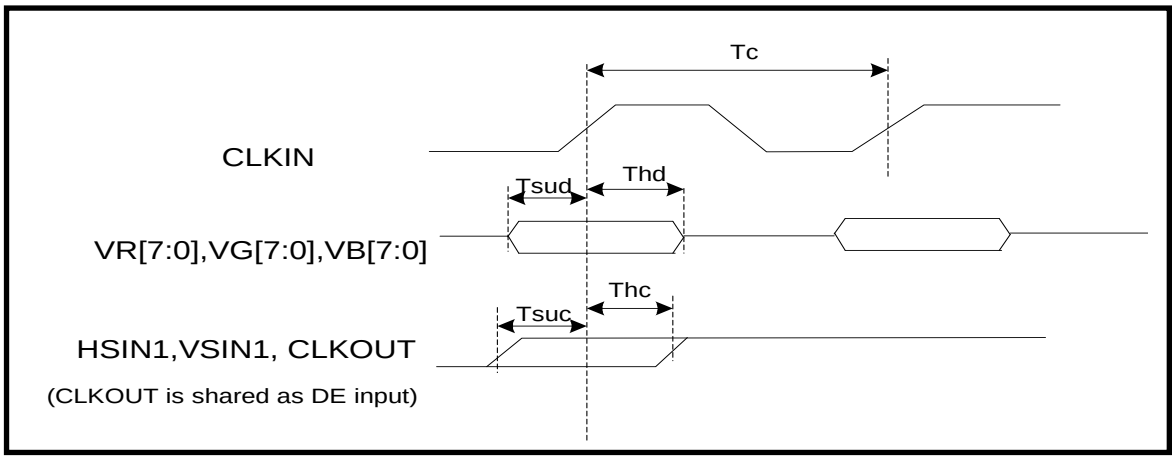


Table 14: Video Input Port AC Timing

Parameter	Symbol	Min	Typ	Max	Units	Conditions
VR[7:0], VG[7:0], VB[7:0] Setup Time to CLKIN	Tsud	2			ns	
VR[7:0], VG[7:0], VB[7:0] Hold Time to CLKIN	Thd	5			ns	
HSIN1, VSIN1, CLKOUT Setup Time to CLKIN	Tsuc	2			ns	
HSIN1, VSIN1, CLKOUT Hold Time to CLKIN	Thc	6			ns	
CLKIN Frequency	1/Tmc			80	MHz	

AC CHARACTERISTICS

图19：视频输入端口交流时序

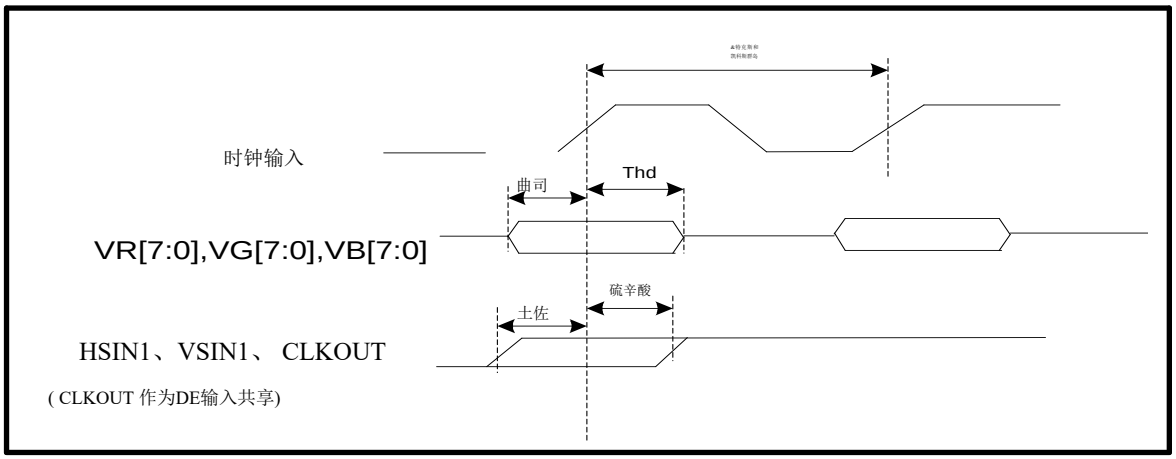


表14：视频输入端口交流时序

参数	符号	微小的	类型	最高的	单位	条件
VR[7:0]、VG[7:0]、VB[7:0] 的 CLKIN 设置时间	曲司	2			神经元	
VR[7:0]、VG[7:0]、VB[7:0] 持续时间至 CLKIN	Thd	5			神经元	
HSIN1、VSIN1、 CLKOUT 设置时间至 CLKIN	土佐	2			神经元	
HSIN1、VSIN1、 CLKOUT 保持时间至 CLKIN	硫辛酸	6			神经元	
CLKIN 频率	1/吨煤			80	兆赫	

Figure 20: Video Output Port AC timing

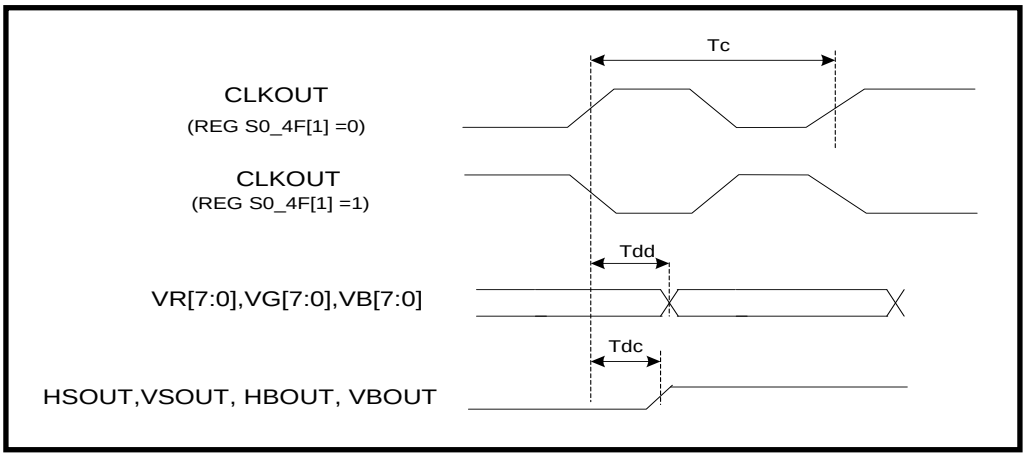


Table 15: Video Output Port AC timing

Parameter	Symbol	Min	Typ	Max	Units	Conditions
VR[7:0], VG[7:0], VB[7:0] Delay Timing from CLKOUT	Tdd	-1		1.5	ns	20pF load
HSOUT, VSOUT, HBOUT, VBOUT Delay Timing from CLKOUT	Thd	-1		2.5	ns	20pF load
CLKOUT Frequency	1/Tmc			108	MHz	20pF load

图20：视频输出端口交流时序

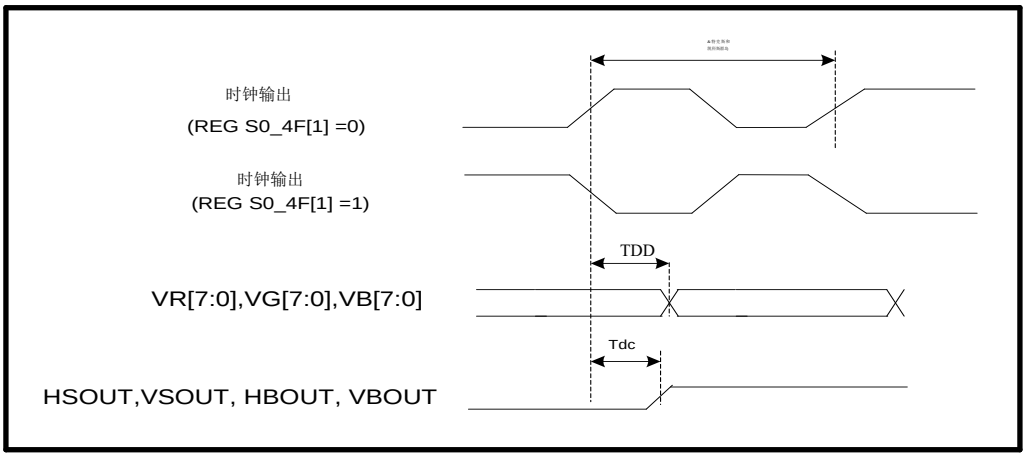


表15：视频输出端口交流时序

参数	符号	微小的	类型	最高的	单位	条件
VR[7:0]、VG[7:0]、VB[7:0] CLKOUT 延迟定时	TDD	-1		1.5	神经元	20皮法负载
HSOUT 、 VSOUT 、 HBOUT 、 VBOUT 延迟定时器 CLKOUT	Thd	-1		2.5	神经元	20皮法负载
CLKOUT 频率	1/吨煤			108	兆赫	20皮法负载

MEMORY INTERFACE AC CHARACTERISTICS

Figure 21: Memory Interface AC Input timing

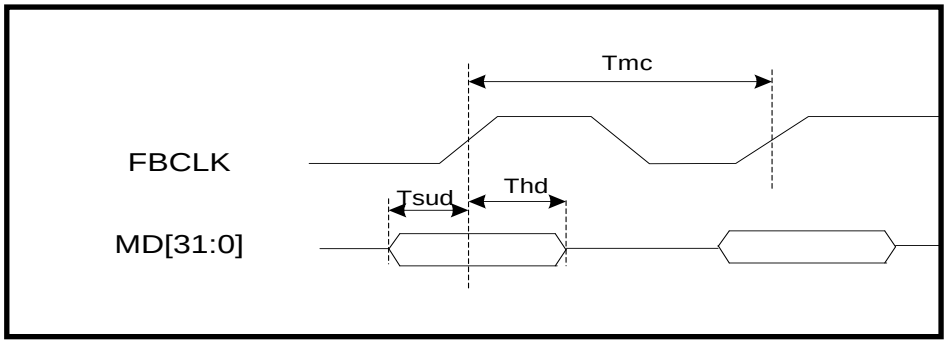


Table 16: Memory Interface AC Input timing

Parameter	Symbol	Min	Typ	Max	Units	Conditions
MD[31:0] Setup Time to FBCLK	Tsud	2.0			ns	
MD[31:0] Hold Time to FBCLK	Thd	1.5			ns	
FBCLK Frequency	1/Tmc			162	MHz	

M 内存 I 接口 AC C 特性

图21：存储器接口交流输入时序

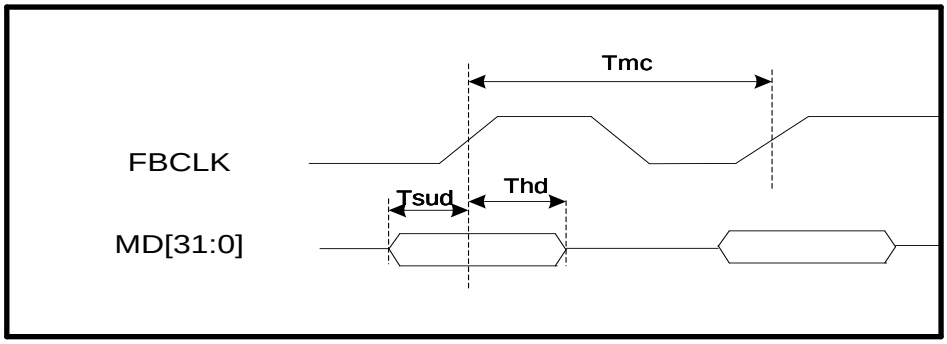


表16：存储器接口AC输入时序

参数	符号	微小的	类型	最高的	单位	条件
MD[31:0] FBCLK 设置时间	曲司	2.0			神经元	
MD[31:0] FBCLK 保持时间	Thd	1.5			神经元	
FBCLK 频率	1/吨煤			162	兆赫	

Figure 22: Memory Interface AC output timing

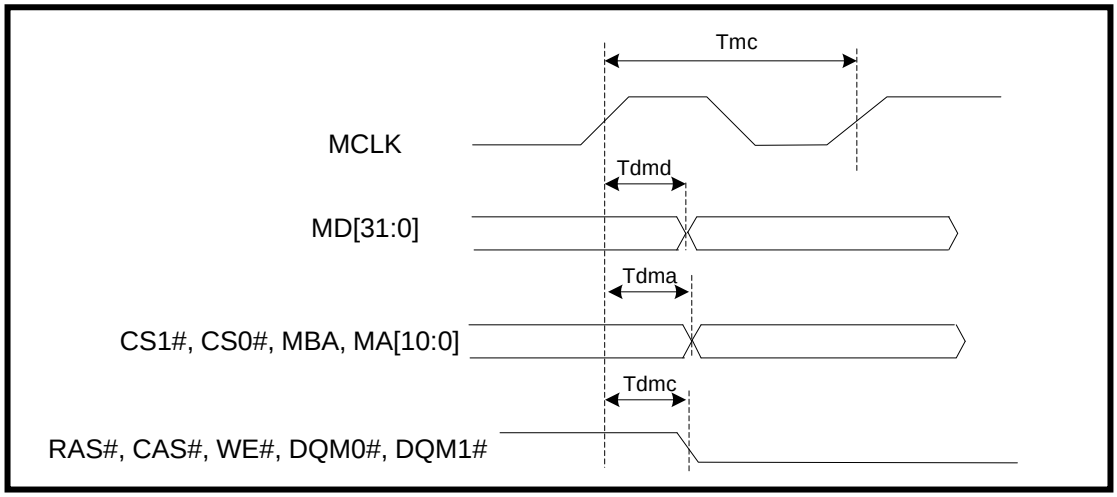


Table 17: Memory Interface AC output timing

Parameter	Symbol	Min	Typ	Max	Units	Conditions
MD[31:0] Delay Time from MCLK	Tdmd	3.7		5.4	ns	20pF load
CS1#, CS0#, MBA, MA[10:0] Delay Time from MCLK	Tdma	3.7		5.4	ns	20pF load
RAS#, CAS#, WE#, DQM0#, DQM1# Delay Time from MCLK	Tdmc	3.7		5.4	ns	20pF load
MCLK Frequency	1/Tmc			162	MHz	20pF load

图22：存储器接口AC输出时序

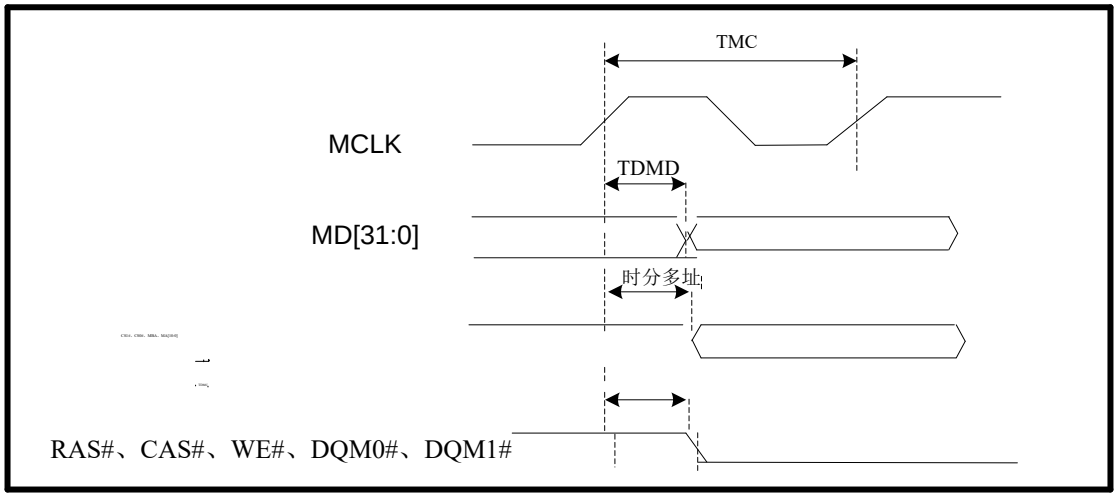
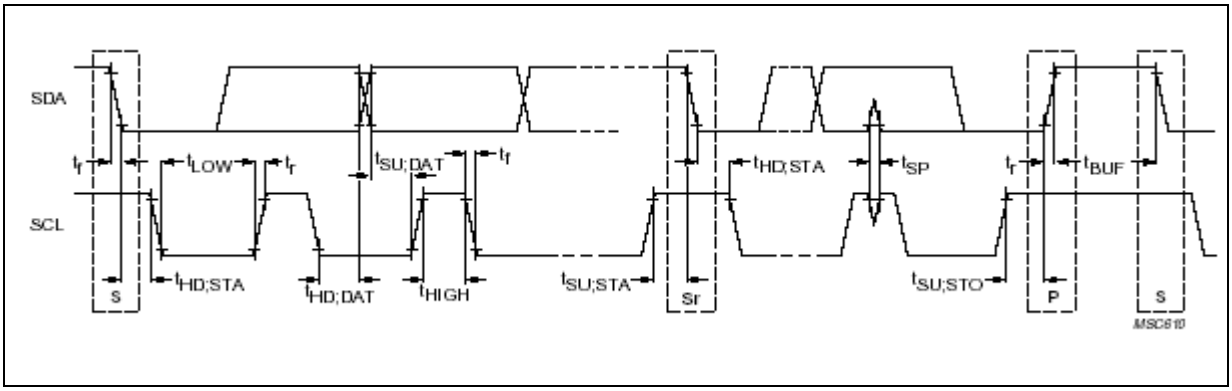


表17：存储器接口AC输出时序

参数	符号	微小的	类型	最高的	单位	条件
MD[31:0] MCLK 延迟时间	TDMD	3.7		5.4	神经元	20皮法负载
CS1#、CS0#、MBA、MA[10:0] MCLK 延迟时间	时分多址	3.7		5.4	神经元	20皮法负载
RAS#、CAS#、WE#、DQM0#、DQM1# MCLK 延迟时间	TDMC	3.7		5.4	神经元	20皮法负载
MCLK 频率	1/吨煤			162	兆赫	20皮法负载

Figure 23: Definition of timing for F/S-mode devices on the I²C-bus

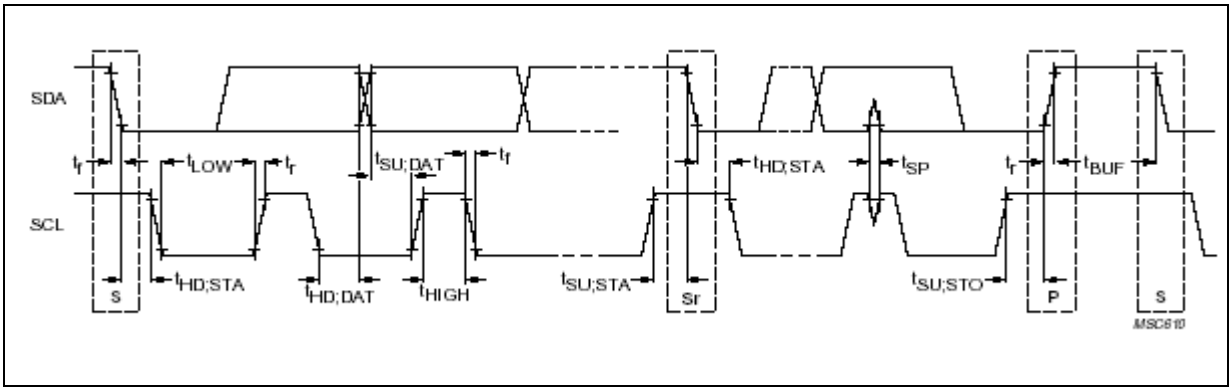


**** Figure 7 Referenced document:***

Philips Semiconductors, "The I²C-BUS Specification, version2.1 ---January 2000"

Page33: Fig.31: Definition of timing for F/S-mode devices on the I²C-bus.

图23：I:C总线F/S模式设备时序定义



** 图7 参考文献:*

飞利浦半导体公司《 I²C 总线规范》2.1版（2000年1月）第33页：图31： I²C 总线F/S模式设备时序定义。

Table 18: Characteristics of the SDA and SCL bus lines for F/S-mode I²C-bus devices

PARAMETER	SYMBOL	STANDARD-MODE		FAST-MODE		UNIT
		MIN.	MAX.	MIN.	MAX.	
SCL clock frequency	f_{SCL}	0	100	0	400	kHz
Hold time (repeated) START condition. After this period, the first clock pulse is generated	$t_{HD;STA}$	4.0	–	0.6	–	µs
LOW period of the SCL clock	t_{LOW}	4.7	–	1.3	–	µs
HIGH period of the SCL clock	t_{HIGH}	4.0	–	0.6	–	µs
Set-up time for a repeated START condition	$t_{SU;STA}$	4.7	–	0.6	–	µs
Data hold time: for CBUS compatible masters for I²C-bus devices	$t_{HD;DAT}$	 5.0 0(2)	 – 3.45(3)	 – 0(2)	 – 0.9(3)	 µs µs
Data set-up time	$t_{SU;DAT}$	250	–	100(4)	–	ns
Rise time of both SDA and SCL signals	t_r	–	1000	$20 + 0.1C_b$(5)	300	ns
Fall time of both SDA and SCL signals	t_f	–	300	$20 + 0.1C_b$(5)	300	ns
Set-up time for STOP condition	$t_{SU;STO}$	4.0	–	0.6	–	µs
Bus free time between a STOP and START condition	t_{BUF}	4.7	–	1.3	–	µs
Capacitive load for each bus line	C_b	–	400	–	400	pF
Noise margin at the LOW level for each connected device (including hysteresis)	V_{nL}	$0.1V_{DD}$	–	$0.1V_{DD}$	–	V
Noise margin at the HIGH level for each connected device (including hysteresis)	V_{nH}	$0.2V_{DD}$	–	$0.2V_{DD}$	–	V

Notes:

1. A device must internally provide a hold time of at least 300 ns for the SDA signal (referred to the V_{IHmin} of the SCL signal) to bridge the undefined region of the falling edge of SCL.
2. The maximum $t_{HD;DAT}$ has only to be met if the device does not stretch the LOW period (t_{LOW}) of the SCL signal.
3. A Fast-mode I²C-bus device can be used in a Standard-mode I²C-bus system, but the requirement $t_{SU;DAT} \geq 250$ ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the SCL signal. If such a device does stretch the LOW period of the SCL signal, it must output the next data bit to the SDA line $t_{r\,max} + t_{SU;DAT} = 1000 + 250 = 1250$ ns (according to the Standard-mode I²C-bus specification) before the SCL line is released.
4. C_b = total capacitance of one bus line in pF.

**** Table 17 Referenced document:***

Philips Semiconductors, "The I²C-BUS Specification, version2.1 ---January 2011"

Page32: Table 5: Characteristics of the SDA and SCL bus lines for F/S-mode I²C-bus devices[1]

表18：F/S模式 I²C 总线设备的SDA与SCL总线线特性

PARAMETER	SYMBOL	STANDARD-MODE		FAST-MODE		UNIT
		MIN.	MAX.	MIN.	MAX.	
SCL clock frequency	f_{SCL}	0	100	0	400	kHz
Hold time (repeated) START condition. After this period, the first clock pulse is generated	$t_{HD;STA}$	4.0	–	0.6	–	µs
LOW period of the SCL clock	t_{LOW}	4.7	–	1.3	–	µs
HIGH period of the SCL clock	t_{HIGH}	4.0	–	0.6	–	µs
Set-up time for a repeated START condition	$t_{SU;STA}$	4.7	–	0.6	–	µs
Data hold time: for CBUS compatible masters for I²C-bus devices	$t_{HD;DAT}$	 5.0 0(2)	 – 3.45(3)	 – 0(2)	 – 0.9(3)	 µs µs
Data set-up time	$t_{SU;DAT}$	250	–	100(4)	–	ns
Rise time of both SDA and SCL signals	t_r	–	1000	$20 + 0.1C_b$(5)	300	ns
Fall time of both SDA and SCL signals	t_f	–	300	$20 + 0.1C_b$(5)	300	ns
Set-up time for STOP condition	$t_{SU;STO}$	4.0	–	0.6	–	µs
Bus free time between a STOP and START condition	t_{BUF}	4.7	–	1.3	–	µs
Capacitive load for each bus line	C_b	–	400	–	400	pF
Noise margin at the LOW level for each connected device (including hysteresis)	V_{nL}	$0.1V_{DD}$	–	$0.1V_{DD}$	–	V
Noise margin at the HIGH level for each connected device (including hysteresis)	V_{nH}	$0.2V_{DD}$	–	$0.2V_{DD}$	–	V

记下

1. 该设备必须在内部为SDA信号（即SCL信号的V_{IHmin}）提供至少300纳秒的保持时间，以桥接SCL信号下降沿的未定义区域。
2. 仅当该装置不拉伸SCL信号的LOW周期（t_{LOW}）时，才需满足最大$t_{HD;DAT}$。
3. 快速模式I²C-总线设备可用于标准模式I²C-总线系统，但需要满足$t_{SU;DAT}$ ≥ 必须满足250纳秒。如果设备不拉伸SCL信号的低电平周期，这种情况将自动实现。如果此类设备确实拉伸了SCL信号的低电平周期，则必须在SCL线释放前将下一个数据位输出到SDA线$t_{r\,max} + t_{SU;DAT}$ = 1000 + 250 = 1250纳秒（根据标准模式I²C -bus规范）。
4. C_b = 一条总线线路的总电容（单位：皮法）。

** 表17 参考文件：*

飞利浦半导体公司《 I²C 总线规范》2.1版（2011年1月）

第32页：表5：F/S模式 I²C 总线 devices[1] 的SDA与SCL总线特性

DAC CHARACTERISTICS

Table 19: DAC Characteristics

Parameter	Min	Typical	Max	Unit	Notes
Resolutions			10	Bits	
Number of channels		4			
Clock Rate		162		MHz	
INL	-1.0		+1.0	LSB	
DNL	-1.0		+1.0	LSB	
Full Scale Voltage	665	700	770	mV	1, 2
LSB Current		18.2		uA	1
Monotonicity		Guaranteed			
RGB Video Output Rise Time (10-90% of full-scale)		2.6	4.7	ns	3
RGB Video Output Fall Time (10-90% of full-scale)		2.5	4.0	ns	3

NOTES:

1. For VESA Video Levels, the R_{iref} in the board is 160ohms. The output load is double terminated with 75ohms and 10pf per channel. Full-scale for all four channels is 0.7V.
2. For good linearity, the full-scale voltage should be less than 1V when the output load is 75ohms and 10pf per channel.
3. As measured with 37.5ohm and 10pf load.

DAC CHARACTERISTICS

表19：DAC特性

参数	微小的	典型的	最高的	单元	记下
决议			10	位	
通道数	4				
时钟速率		162		兆赫	
INL	-1.0		+1.0	最低效位元	
DNL	-1.0		+1.0	最低效位元	
满量程电压	665	700	770	马尔代夫	1, 2
LSB 电流		18.2		uA	1
单调性	有保证的				
RGB视频输出上升时间（10-90%的全尺寸）		2.6	4.7	神经元	3
RGB视频输出的延迟时间（10-90%的全尺寸）		2.5	4.0	神经元	3

注释：

1. 对于VESA视频电平，电路板中的参考电阻R_{iref}为160欧姆。输出负载采用双端接方式，每通道75欧姆和10皮法。所有四个通道的满量程电压为0.7伏。
2. 为确保良好的线性度，当输出负载为75欧姆且每通道10皮法时，满量程电压应小于1伏。
3. 以37.5欧姆和10皮法负载测量。

ADC CHARACTERISTICS

Table 20: ADC Characteristics

Parameter	Min	Typical	Max	Unit	Notes
Resolutions			8	Bits	
Number of channels		3			
Sampling Frequency (Fs)	10		162	MHz	
Full Scale Adjust Range at RGB Inputs	0.5		1	V	1
Reference Voltage		1.24		V	2
INL		+/- 1.2		LSB	
DNL		+/- 0.8		LSB	
No Missing Codes		Guaranteed			

NOTES:

1. Operation with 8-bit gain controller and 7-bit offset controller.
2. Internal reference voltage output.

ADC C 特性

表20：ADC特性

参数	微小的	典型的	最高的	单元	记下
决议			8	位	
通道数	3				
采样频率（Fs）	10		162	兆赫	
RGB输入的全标度调整范围	0.5		1	V	1
参考电压		1.24		V	2
INL		+/- 1.2		最低效位元	
DNL		+/- 0.8		最低效位元	
无缺失代码	有保证的				

注释：

1. 采用8位增益控制器和7位偏移控制器进行操作。
2. 内部参考电压输出

PACKAGE DIMENSIONS

In Millimeters

	Min	Typ	Max
A1	0.25		
A2	3.2	3.32	3.6
L1	1.6REF		
b	0.22	0.3	0.38
c	0.09	0.15	0.2
e	0.65 BSC		
D1		28	
E1		28	

Figure 24: Package Dimensions for QFP160 Package

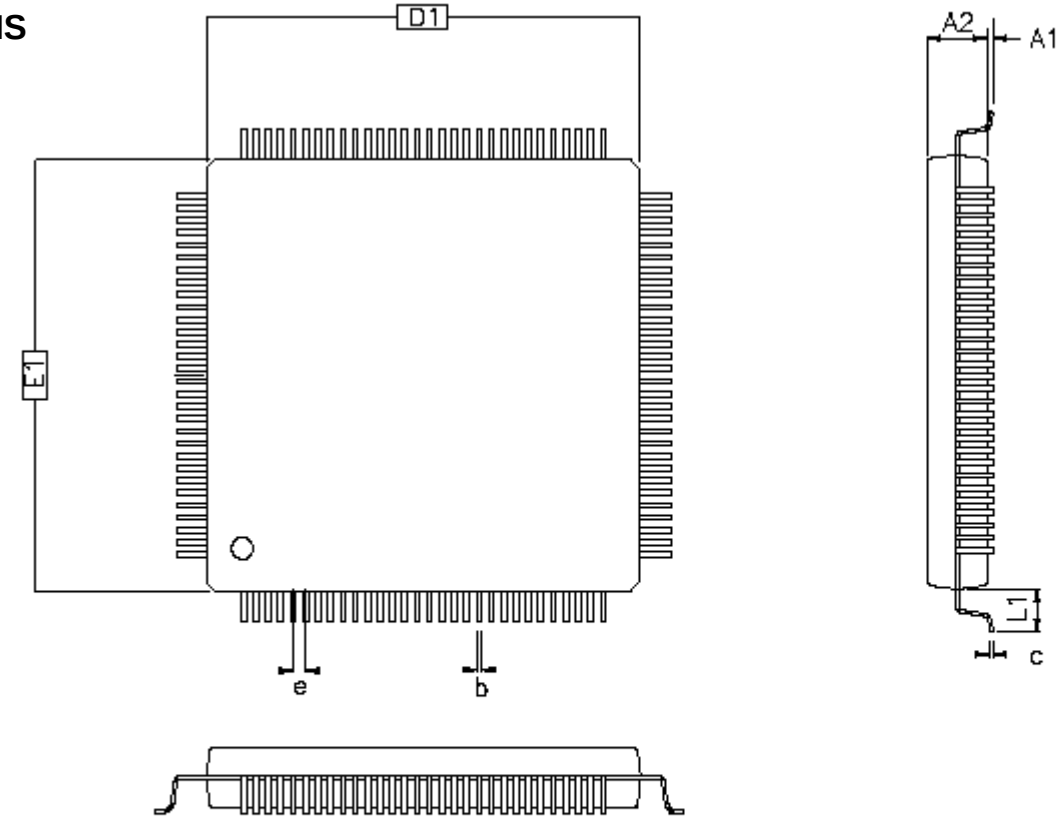
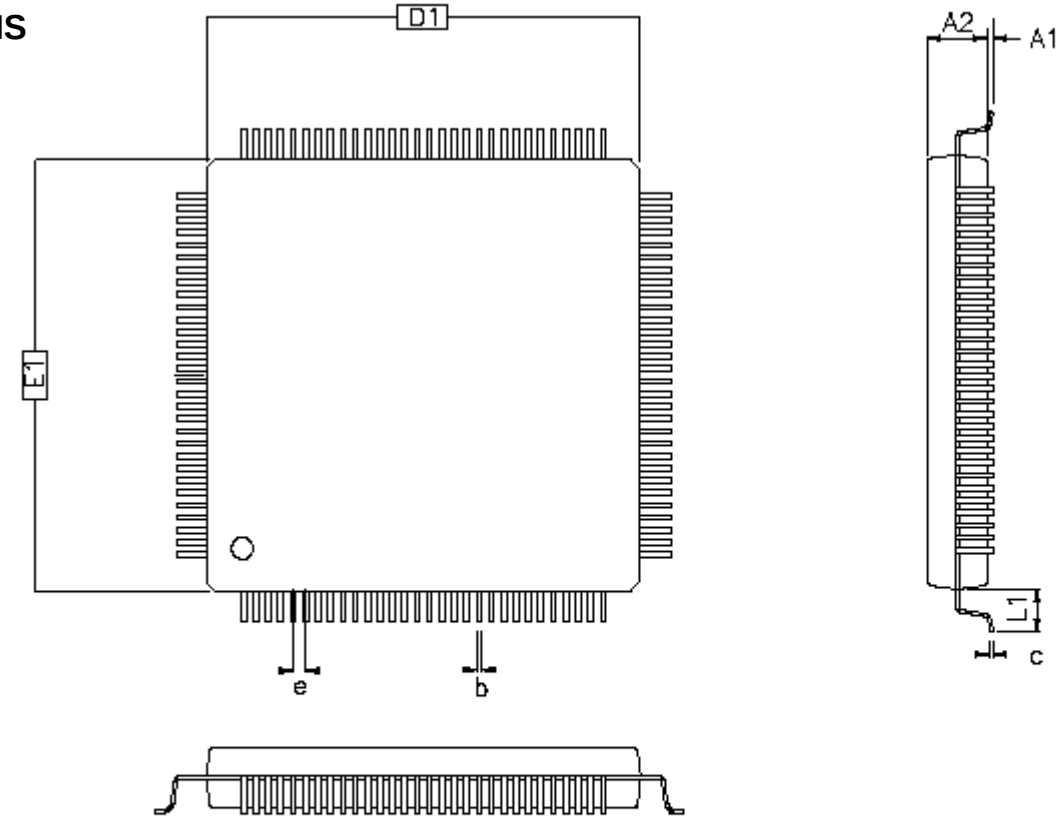


图24：QFP160封装尺寸

PACKAGE DIMENSIONS

以毫米为单位

	微小的	类型	最高的
A1	0.25		
A2	3.2	3.32	3.6
L1	1.6REF		
b	0.22	0.3	0.38
c	0.09	0.15	0.2
e	0.65 基本的		
D1		28	
E1		28	



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